



Uworld review - NCLEX

The Nurse As Intra/Entrepreneur And Consultant (Drexel University)



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UWORLD REVIEW

- TB and all airborne transmission based precautions (chickenpox, measles)
 - Use a surgical mask not an N95 for clients.
 - This helps protect Health care workers and other clients from respiratory secretions, and contain respiratory secretions
 - The HCWs who transport the clients from a negative pressure room to another location need to wear an N95 mask to protect themselves from infection.
- Cirrhosis of the liver. These blood tests would be elevated in the lab results:
 - Ammonia, bilirubin, prothrombin time
 - Elevated bilirubin (jaundice) exists due to the decreased inability of the liver to conjugate and excrete bilirubin
 - Most coagulation factors are produced in the liver. A cirrhotic liver cannot produce the factors essential for blood clotting. As a result, the PT, aPTT, and INR levels are elevated.
- Total knee replacement
 - A recent/current infection is a contraindication to total replacement surgery as a wound infection is more likely to occur in a client with a pre-existing infection.
 - Burning on urination should be investigated as it could indicate a UTI
 - Severe knee pain is expected.
 - *Stop taking NSAIDS including selective COX-2 inhibitors (celecoxib) 7 days prior to surgery to decrease the risk of intra and post op bleeding.*
- Transfusion reaction
 - Chills, fever, low back pain, flushing, itching)
 - Stop transfusion immediately, maintain IV line with NS, monitor VS, notify HCP and blood bank, recheck tags numbers and client's blood type, and return bag to blood bank for further testing. Collect blood and urine sample to evaluate hemolysis, and complete necessary facility paperwork to document the reaction.
- Chronic kidney disease with a large plural effusion. Findings :
 - Decreased fremitus
 - Diminished lung sounds
 - A pleural effusion is an abnormal collection of fluid (>15-20ml) in the pleural space between the parietal and visceral pleurae that prevents the lung from expanding fully. This results in decreased lung volume, Atelectasis, and ineffective gas exchange.
 - Other s/s: dyspnea on exertion, non-productive cough, diminished breath sounds, dullness to percussion, and decreased tactile fremitus. If the effusion is large, the trachea is deviated to the opposite side.
 - Sounds travel faster in solids (consolidation) than in an aerated lung, resulting in increased fremitus in pneumonia. fluid or air outside the lung interrupts the transmission of sound, resulting in decreased fremitus in pleural effusion and Pneumothorax.

- Cancer
 - Unintentional weight loss of greater than 10 percent of usual weight (in nonobese) clients require evaluation and could indicate underlying cancer. N, anorexia, and dysgeusia (altered taste sensation) are also clinical features of cancer and contribute to weight loss.
 - CAUTION acronym of cancer
 - Change in bowel or bladder habits
 - A sore throat that does not heal
 - Unusual bleeding or discharge from body orifice
 - Thickening or lump in the breast or elsewhere
 - Indigestion or difficulty in swallowing
 - Obvious change in wart or mole
 - Nagging cough or hoarseness
 - Orange peel appearance of breast tissue, or retracted nipple
- Restrained client
 - Position sideways or semi fowlers, because supine position can cause risk of aspiration
 - Don't tie the knot a square knot. Do a quick release knot.
 - Tie to bed frame not bed rail
 - Provide care to meet basic needs every 2 hours
- Gout
 - To prevent future exacerbations
 - 1) Achieve and maintain a healthy weight
 - 2) Drink plenty of fluids
 - 3) Limit alcohol consumption and carbonated beverages containing high fructose corn syrup.
 - You don't need to eliminate all foods with protein. Just red meats and seafood intake
- Hip fractures
 - external rotation, abduction, muscle spasm, and shortening of the affected extremity.
- Raynaud's phenomenon
 - An episodic vasospastic disorder of the small cutaneous arteries, mainly of the fingers and toes.
 - Occurs most often in young women age 15-40.
 - Vasospasm induced color changes of the fingers, toes, ears, and nose.
 - Decreased perfusion initially causes pallor (white) followed by cyanotic (blue purple) and last color change is red.
 - Coldness numbness, followed by throbbing, aching, tingling, swelling (hyperemic phase).
 - Exposure to cold, emotional stress, and caffeine, and tobacco use may bring on symptoms.
 - Don't use substances with vasoconstrictive properties

- Perform stress management
- Emphysema
 - Barrel chest, decreased activity tolerance, diminished breath sounds, distant heart sounds
 - Wheezing
 - SOB that worsens over time
 - tachypnea, prolonged expiratory phase
 - Hyperresonance on percussion, pursed-lip breathing, use of accessory muscles and use of tripod position
- Hyperosmolar hyperglycemic state complications:
 - Blood glucose levels are very high (>600)
 - History of type 2 diabetes. Older age
 - Neurological manifestations (blurry vision, lethargy, obtundation (decreased LOC), progression to coma)
 - Gradual onset of hyperglycemic symptoms (as some insulin is produced)
 - Hyperventilation and abdominal pain less common
 - Bicarb greater than 18 mEq/L
 - Normal anion gap
 - Serum osmolality >320 mOsm/kg
 - **Negative or small serum ketones**
- Diabetes insipidus
 - Insufficient production or suppression of ADH
 - Polydipsia (increased thirst); polyuria (increased urine output) and can lead to dehydration resulting in weight loss, hypernatremia, and high serum osmolality (>295 mmol/kg).
 - Dilute and copious urine (2-20 L/day) with a low specific gravity (<1.005)
- **Dilantin (Phenytoin)**
 - Anticonvulsant used to treat generalized tonic-clonic seizures.
 - Therapeutic range: 10-20 mcg/ml, and it takes 3-12 months for steady state to be reached.
 - Early toxicity signs: horizontal nystagmus and gait unsteadiness. Slurred speech, lethargy, confusion, and even coma. Bradyarrhythmias and hypotension are usually seen with IV Phenytoin
 - Gingival hyperplasia is common expected side effect of Phenytoin and does not indicate drug toxicity. Occurs more in those less than 23 y/o who are prescribed >500 mf/day. Good oral hygiene can limit symptoms.
 - Other s/s: ataxia, decreased alertness
- **Metronidazole**
 - Metallic taste in the mouth is common

- It's an antibiotic
- Trisomy 18
 - A genetic disorder with a short life expectancy (a few weeks after birth) . It's a chromosome anomaly characterized by severe cardiac defects and multiple musculoskeletal deformities.
 - End of life issues should be discussed early after the diagnosis is confirmed. Trisomy 13 (patau syndrome) also results in death
 - No treatment at this time available
- Warfarin
 - Anticoagulant. Range: 2.3-3.5.
 - Warfarin should only be administered after INR has been checked. Can be given if it is less than 3.5 and should be held and HCP contacted if greater than 3.5.
 - Antidote: Vitamin K
- Neutrophils
 - Normal: >1500/mm³
 - Neutropenic precautions: private room, strict Handwashing, avoid exposure to people who are sick, avoid ALL fresh fruits, veggies, and flowers; ensure all equipment used with the client has been disinfected.
 - Infections in immunosuppressed clients are life threatening
 - Avoid IM injections and minimize venipunctures when platelet count is below 50,000 as these can cause prolonged bleeding
- Blood lead levels
 - Common source is lead based paint found in houses built before 1978.
 - Screenings are recommended at ages 1, 2, and up to 6, if not previously tested.
 - Lead poisoning particularly affects the neurological system, due to immature development of the brain and nervous system. This is more concerning than the other effects.
 - Can cause neuro impairment, developmental delays, reading difficulties, visual-motor issues; can lead to permanent cognitive impairment, seizures, blindness, or even death.
 - GI bleeding is for iron toxicity not lead poisoning.
 - Lead poisoning is the most threatening to the kidneys and neuro system (developmental delays, cognitive impairment, seizures).
 - Liver injury typically does not occur. Severe liver damage is closely associated with acetaminophen overdose or Reye syndrome.
- Asystole
 - Total absence of ventricular electrical activity (pulseless, apneic, unresponsive)
 - Treatment: CPR, ACLS, epinephrine and/or vasopressin, advanced airway use, and any reversible treatment.
 - Vasopressin is a vasopressor, that increases vasoconstriction and MAP

- SIDS
 - Sudden unexplained death of an infant age <1 year. Highest occurrence at 2-4 months.
 - Protective factors:
 - Supine sleeping position
 - Breast feeding
 - Pacifier use during sleep
 - Up to date vaccinations
 - Appropriate clothing (sleep sack)
 - Firm sleep surface
 - Removal of loose items from bed
 - Risk factors:
 - Prone/side sleeping position
 - Tobacco smoke exposure before/after birth or any drugs
 - Premature birth/low birth weight
 - Overheating during sleep
 - Soft sleep surface
 - Bumper pads, toys and loose pillows/blankets
- Adalimumab (tumor necrosis factor) inhibitor
 - Reduces inflammatory response for rheumatoid arthritis. TNF is a factor in the immune system response that increases inflammation.
 - Cause immunosuppression, and increases risk of new infections or reactivation of previous infections (latent TB, hepatitis B virus)
 - Other TNF inhibitors: etanercept, infliximab
- **Opioid** intoxication clinical features:
 - Depressed mental status
 - **Decreased resp rate (<12/min) most notable**
 - Constricted (miotic) pupils not present in every client
 - Decreased/absent bowel sounds
 - Mild hypotension from histamine release and Bradycardia from CNS depression
 - Taking other CNS depressants like alcohol and benzo can worsen resp depression
 - **Flumazenil is a benzo antagonist that can be administered if there is no response to the naloxone (Opioid antagonist) and ingestion of benzo is suspected.**
 - **Administer naloxone (Narcan)** a potent narcotic antagonist to reverse CNS and resp depression
- Burn patients
 - Initial management is ABCs
 - Should be treated with high flow O2 via a non-rebreather mask initially.
 - Requires significant volume replacement to compensate for fluids lost through wounds in more than 15% total body surface area (TBSA) involved.
 - Lactated ringers usually used. Follow the Parkland formula (4 mg/kg of body weight

- for each percent of tbsa burned). $\frac{1}{2}$ of the total amount is infused in the first 8 hours; $\frac{1}{4}$ of the total in the 2nd 8 hours, and $\frac{1}{4}$ of the total in the 3rd 8 hours.
- Full thickness burns involve destruction of nerves, so there is no pain
- **Case manager role**
 - Assessing, planning, facilitating, and advocating for client health services to accomplish cost-effective quality client outcomes. This is done through communication and use of available resources. A professional nurse often serves in the case manager role. Assesses client needs, decreases fragmentation of care; helps to coordinate care and communication between HCPs, make referrals, ensure quality standards are being met, and arranges for home health or placement after discharge..
 - They do not typically provide direct client care. Medication reconciliation should be done between primary nurse and the HCP.
 - They usually make daily rounds to the nursing dept. To review documentation in client charts but not necessarily visit the client personally.
 - **Ziprasidone HCL (Geodon)**
 - Atypical antipsychotic drug that is used for acute bipolar mania, acute psychosis, and agitation.
 - **Risk: QT prolongation leading to torsade de pointes**
 - Monitor for: hypotension, seizures.
 - A baseline ECG and K are usually checked. At a minimum, the client should be placed on a cardiac monitor.
 - Alcohol interacts with the drug and increases the potential for an adverse effect from the drug.
 - **Isotretinoin (accutane)**
 - Decreases sebum secretion and is prescribed for severe, disfiguring, nodular acne, that has been unresponsive to other therapies, including antibiotics.
 - **Pregnancy category X** drug and is known to cause severe birth defects if taken during pregnancy.
 - Females must have 2 pregnancy tests before taking the med. Also, 2 forms of contraception must have been in place for at least 1 month prior to starting this med.
 - Before refills can be obtained, enrollment in a risk management program is required to verify that pregnancy tests are negative and 2 forms of contraception are being used.
 - Blood donation is also discouraged while on therapy and for 1 month after to ensure that pregnant women do not receive any donated blood.
 - Dryness of the eyes, mouth, and skin are common s/e. Lubricating eye drops may be needed to wear contacts. Some clients are unable to wear contacts while taking this medication. Good oral hygiene and skin care are needed.
 - Capsules should be swallowed whole with at least 8 oz water or other fluid. It should not be broken, crushed, or chewed as contents of opened capsules could irritate esophagus.
 - This causes photosensitivity. Use sunscreen routinely.

- **TB drugs**

- Isoniazid, rifampin, ethambutol, pyrazinamide
- Isoniazid causes hepatotoxicity and peripheral neuropathy.
- Rifampin causes hepatotoxicity.
- Baseline liver function tests should be obtained.
- Watch for s/s of hepatotoxicity (jaundice, anorexia).
- Ethambutol causes ocular toxicity, and clients will need frequent eye examination.
- Rifampin:
Changes color of body fluids (urine, sweat) due to its body wide distribution (red-orange colors). Tears can turn red, making contact lenses appear discolored. Client should wear sunglasses instead of soft contact lenses while taking this medication. Women should use non hormonal birth control methods while taking this drug as it can decrease the effectiveness of oral contraceptives.
- Don't consume alcohol and drugs that can increase the risk for hepatotoxicity (acetaminophen) during long term use of this drug.
- The effectiveness of treatment for active TB is determined by 3 negative sputum cultures and chest x-ray. If the entire course of therapy (6-9 months) is not completed, re infection, spread to others, and development of resistant strains of TB bacteria can result).
- **Increased metabolism of some drugs (oral contraceptives, hypoglycemic, and warfarin) can occur as s/e/**

- Vitamin K is the antidote for warfarin related bleeding.
- aPTT 46-70, INR 2-3
- **Sodium polystyrene sulfonate (kayexalate) is a sodium exchange resin administered to reduce elevated serum K levels in clients with chronic kidney disease and hyperkalemia.**
- Warfarin is started about 5 days before a continuous heparin infusion is d/c as it takes this long for warfarin to reach therapeutic levels

- **Propranolol**

- Beta blocker that inhibits beta1 (heart) and beta2 (bronchial) receptors. It is used for many indications (essential tremor) in addition to BP control. BP decreases secondary to a decrease in HR.
- Bronchoconstriction may occur due to the effect on the B2 receptors.
- Wheezing in a client taking this drug may indicate bronchoconstriction or bronchospasm. **Nurse should assess for any history of asthma or resp problems with this client and notify the HCP.**
- Headache is common occurrence with HTN. Analgesic can be given as needed.
- It may take several days of treatment for the BP to reduce to a more normal finding.

Reduction in HR is expected with beta-blockers. Nurse should monitor it for further reduction.

- **Pancreatitis**

- Pancreatic enzymes help with digestion and are inactive when they are inside the pancreas. They activate in the small intestine.
- If you are an alcoholic scar tissue occurs in pancreas. It creates an occlusion in pancreas and prevents enzymes coming out
- Gallstones can occlude the enzymes from coming out too
- The enzymes then activate inside the pancreas as it gets tired to wait to activate. This causes pancreatitis (pancreas eats itself)

Hepatic encephalopathy

- Serious complication of end stage liver disease resulting from inadequate detoxification of ammonia from the blood. (lethargy, confusion, slurred speech)
- Increased ammonia levels and asterixis are characteristic of HE.
- **Asterixis (flapping tremors of the hands)**. It is assessed by having the client extend the arms and Dorsiflex the wrists.
- Feter hepaticus (musty, sweet odor of breath) from accumulated digestive by products.
- Other s/s: sleep disturbances, lethargy, and altered mental status, coma (decreased LOC)
- Spider angiomas, gynecomastia, testicular atrophy, and palmar erythema are expected findings in cirrhosis d/t altered metabolism of hormones in liver.
- Precipitating factors: hypokalemia, constipation, GI hemorrhage, infection
- It's the frequent complication of liver cirrhosis. It results from accumulation of ammonia and other toxic substances in the blood.
- **Lactulose is the DOC which helps excrete ammonia through the bowels as soft or loose stools. Rifaximin (antibiotic) can also be given**
- Clients with cirrhosis typically have hypokalemia due to hyperaldosteronism (as it is not metabolized by the damaged liver). It can also occur d/t diuretics used to treat fluid retention and ascites.

Ileostomy

- Surgical procedure that creates an opening (stoma) in the abdominal wall that originated from the intestinal mucosa of the ileum.
- In the beginning: LOW FIBER DIET. To prevent obstruction of the narrow lumen of the small intestine.
- Foods to be avoided include:
 - Stringy texture: celery, broccoli, asparagus stalks
 - High fiber: bread, brown rice, oatmeal
 - Seeds or pits: strawberries, raspberries, olives
 - Edible peels: apple slices, raw cucumber, dried fruits
 - Nuts: peanuts, almonds, cashews
- Foods to eat:
 - Fruits and veggies that are pitted, peeled cooked
 - Low fiber carbs (white rice, refined grains/pastas, cooked cereals)

Dumping syndrome

- Rapid emptying of hypertonic gastric contents into duodenum and small intestine

- This leads to fluid shift from the Intra-vascular space to the small intestine, leading to hypotension and activation of the SNS. (abdo pain, diarrhea, N, V, dizziness, general sweating, tachycardia-palpitations)
- Need small frequent meals, foods high in protein and fats, drink fluids between meals, avoid meals high in carbs; have diets high in fiber; eat slowly; avoid sitting up after a meal

Refeeding syndrome

- Serious complication of nutritional replenishment
- **Marked by declines in serum Phosphorus, potassium, and/or magnesium (PPM)**
- Clients can also develop fluid overload
- A low calorie feeding and gradual increase in calories can prevent refeeding syndrome.
- Electrolytes should be monitored carefully and obtain baseline electrolytes.

Dicuclomine hydrochloride (bentyl)

- Anticholinergic used to relax smooth muscle and dry secretions.
- s/e: papillary dilation, dry mouth, urinary retention, and constipation
- classic contraindications: closed – angle glaucoma, bowel ileus, and urinary retention (also BPH).
- Other contraindications: in the presence of bowel ileus, or atony, as constipation is a s/e and further relaxation of the intestines could worsen these conditions.
- is used to treat a certain type of intestinal problem called irritable bowel syndrome. It helps to reduce the symptoms of stomach and intestinal cramping. **This medication works by slowing the natural movements of the gut and by relaxing the muscles in the stomach and intestines.**
- **antispasmodic**

Cholecystectomy

- After the procedure: place in SIMS position. It helps prevent respiratory complications. It helps with the movement of CO₂ used during surgery to fill the abdominal cavity. CO₂ can irritate the phrenic nerve and diaphragm, causing breathing difficulty.
- Diet : related to weight reduction and maintaining a low fat diet.

SIRS

- Systemic inflammatory response syndrome (sepsis)
- **A complication of pneumonia**

LAB VALUES

WBC count: 4,000 -11,000/mm³

BUN levels: 7-18 mg/dL

Creatinine level: 0.6-1.2 md/dL

Hemoglobin levels: 13.5-17 g/dL in men; 12-16 in women).

Magnesium levels: 1.5-2.5 mEq/L

Salicylate acid (aspirin)

- Toxicity: tinnitus, dizziness, N, V, hypotension, tachycardia, **tachypnea, hyperventilation,** and decreased LOC.
- Accumulation of acids results in metabolic acidosis. Respiratory system compensates for increased metabolic acidity by hyperventilating. This is the body's attempt to restore acid-base balance by blowing off carbon dioxide (acid gas) to normalize the pH.

Pyelonephritis

- UTIs that occur in the kidneys (inflammation of the kidney parenchyma) (in the bladder-cystitis) and or urethra (urethritis).
- It causes flank pain that is experienced in the back at the costovertebral angle and may spread to toward the umbilicus.
- Cystitis alone does not cause N, V, or chills. Presence of these, fever, and signs and symptoms of a lower UTI (Dysuria, urgency, and frequency) indicate pyelo...
- Pain in pyelo is dull, constant, and maximal at the costovertebral angle area.

Cystitis

- UTI in bladder
- Suprapubic pain, and spasms, dull and continuous

Distended bladder

- Constant pain increased by pressure over the bladder.
- Distention is found through palpation (firmness, pain, urgency) and percussion (dullness) over suprapubic area.

Renal Colic pain

- Excruciating, sharp, stabbing
- Unable to find a comfortable position; tossing in bed
- Pain radiates down to the groin area from flank, as stone travels down the ureter.

Acute Rheumatic Fever

- An acute inflammatory disease of the heart. It occurs about 2-3 weeks after a streptococcal pharyngitis
- RF affects the heart, skin, joints, and CNS. The presence of 2 major or 1 major and 2 minor criteria and evidence of a preceding streptococcal infection indicates a high probability of RF.
- Family history, and failing to complete a course of Abs is not a direct cause or risk of RF. Fever is a symptom of many illnesses, not just RF.
- The nurse should ask about a streptococcal throat infection when collecting health history info in a client suspected of having RF.

Infection Transmission:

Airborn: TB, influenza (airborne droplets)

Pneumonia: respiratory disease

Sepsis

- SIRS (systemic inflammatory response syndrome) with an infection (suspected or proven). (complication of pneumonia)
- Can be diagnosed with 2 of these 4 criteria:
Temp: $>38^{\circ}\text{C}$ or $<36^{\circ}\text{C}$
HR: >90 bpm
R: $>20/\text{min}$
WBC $>12,000/\text{mm}^3$; $<4,000/\text{mm}^3$ or $>10\%$ immature (band) forms
- Additional s/s:
SBP: <90 mm Hg
Altered mental status
Hyperglycemia (>140 mg/dL)
- Early therapy: aggressive fluid resuscitation and early administration of antibiotics.
- Can occur as a complication of pneumonia who don't respond to a/b care. It is caused by the entry of bacteria from the alveoli into the bloodstream. It can progress to septic shock and/or multisystem organ dysfunction syndrome. To limit progression assess oxygenation (pulse oximeter, ABGs); airway (patency); breathing (resp pattern and rate); circulation (vital signs); tissue perfusion (LOC, cap refill, skin temp and color, bowel sounds) and urine output.
- Paralytic ileus occurs in the presence of sepsis and hypoxia as blood is shunted away from the GI to the vital organs.
- Prolonged cap refill ($>3-4$ seconds in an adult) indicates inadequate blood flow to peripheral tissues.
- Serum glucose >140 : gluconeogenesis occurs in response to the physiologic stress of infection. insulin resistance is associated with anaerobic metabolism

Kidney Biopsy

- Bleeding is a major complication following the procedure.
- Pre procedure: client must give consent and d/c all anticoagulants (heparin, warfarin, rivaroxaban) and antiplatelet agents (aspirin, clopidogrel, NSAIDS) for at least 1 week
- Post procedure: monitor v/s at least Q15 mins for the first hour as tachycardia, tachypnea, and hypotension can indicate blood loss. Assess puncture site dressing for bleeding.
- BUN and creatinine levels would not change much within 30-60 mins. These are usually measured once every 24 hours and rarely every 12 hours.
- Inserting an indwelling urinary catheter is not necessary to perform a kidney biopsy.
- Place prone during procedure to facilitate access to kidney
- Place on affected side after procedure to provide pressure and help prevent bleeding.
- Bed rest for 24 hours

Statins

- Prior to starting therapy with statin drugs, the client's liver function tests should be assessed. The drug is metabolized by the hepatic enzyme system and could cause drug induced hepatitis and increased liver enzymes.
- Drug can also cause muscle aches, and rarely severe muscle injury (rhabdomyolysis). Educate client to report the development of muscle pain while on therapy. Assessing muscle strength or dietary intake prior to therapy is not necessary.
- May slightly increase serum digoxin levels, but is not essential to determine this before therapy is started.

Acute urinary retention

- Treated with rapid, complete bladder decompression instead of intermittent urine drainage.
- This can lead to Hematuria, hypotension, and diuresis.
- Maintaining perfusion and adequate blood pressure is the priority concern.
- Bradycardia is also a complication

Pessary is a vaginal support device recommended for pelvic organ prolapse.

Chronic Kidney Disease

- Decreased glomerular filtration rate, resulting in fluid, K, and phosphorus retention. Fluid retention is initially treated with Na restriction and diuretic therapy. Dietary adjustments should also be made to reduce serum K and phos levels.
- Dairy products (milk, yogurt) and certain fruits (banana, oranges, coconuts, watermelons, and avocados) contain high K.

- Allowable foods: apples, pears, grapes, pineapple, blackberries, blueberries, plum

Lasix(furosemide)

- **Potassium depleting loop diuretic.**
- Hypokalemia can lead to heart palpitations and/or dysrhythmias.

Bladder cancer:

- Tell tale symptom: painless Hematuria.
- **Primary cause is cigarette SMOKING or other tobacco use**
- Occupation cal carcinogen exposure is the second most common factor (printing, ironing, and most aluminum processing, industrial painting, metal work, machining, and mining). Clients are exposed to carcinogens through direct skin contact and inhalation (aerosols and vapors)
- Other risk factors: high fat diet and artificial sweeteners

Influenza

- Has an incubation period of 1-4 days, with peak transmission starting at about 1 day before symptoms appear and lasting up to 5-7 days after illness stage begins.
- Transferred through **inhaling airborne droplets** (sneezing, coughing, speaking) and **DIRECT CONTACT**
- Wear a mask if contact with infected person is unavoidable.
- Those with influenza cannot transmit the virus during the incubation period and illness stage of the infection.
- Avoid close contact with others during illness stage, esp those with impaired immune system

TB

- Gram positive acid fast bacillus (mycobacterium tb)
- **Transmitted airborne droplets (cough or sneeze in the air, or exhaling- breathing, singling talking, laughing)**
- **Need standard and airborne transmission precautions and wear a high-efficacy particulate or N95 resp. Mask.**
- **PTB Not spread with contact of clients blood, urine or soiled clothing, bed linens, or eating utensils**
- 85% pulmonary, but can also be extra pulmonary (meninges, genitourinary, bone and joints, GI)
- Symptoms:
Low grade fever

Night sweats

Anorexia and weight loss

Fatigue

- Pulmonary TB: cough, purulent or blood tinged sputum, SOB
- Genitourinary TB: Dysuria
- Liver involvement jaundice, but also as a side effect to TB drugs (Isoniazid)
- Spinal tb: back pain
- Cardinal (major) constitutional (minor) s/s/
- Dyspnea and hemoptysis seen in later stages
- Classic signs of TB can be absent in immunocompromised clients and the elderly.

Vancomycin to an MRSA infected client

- Check the Bun and creatinine before administering the drug (2-3 times a week) d/t increased risk of nephrotoxicity especially in those that are >60 y/o and have an impaired renal function.
- **Vancomycin is excreted by the kidneys.** It is used to treat gram positive (MRS and diarrhea associated with C.Diff).
- **Normal BUN levels: 7-18 mg/dL**
- **Normal creatinine level: 0.6-1.2 md/dL**
- An increased glucose level is expected in clients with an infection due to physiological stress and gluconeogenesis. Does not need to be reported to the HCP.
- An elevated WBC count is expected and not needed to be reported.
- **Hemoglobin levels: 13.5-17 g/dL in men; 12-16 in women).**
- **Magnesium levels: 1.5-2.5 mEq/L**

Antiplatelet therapy (aspirin, clopidogrel, prasugrel, ticagrelor)

- Initiated to prevent platelet aggregation in those at risk for MI, stroke, or other thrombolytic events.
- This therapy increases the risk of bleeding.
- Assess for bruising, tarry stools, and other signs of bleeding. It can cause thrombotic thrombocytopenia purpura.
- Baseline liver enzymes assessment is not needed

Statins and Isoniazid (for TB)

- Assess baseline liver enzymes

Cystoscopy

- Pink tinged urine, frequency, and Dysuria are expected for up to 48 hours following a cystoscopy.
- Need to increase fluid intake (drink 4-6 glasses of water to help dilute urine)
- Avoid alcohol and coffee for 24-48 hours as this can irritate the bladder
- Abdominal discomfort and bladder spasms may occur for up to 48 hours following procedure.
- Frequent complications: retention, hemorrhage, infection. Notify the doctor if there is any bright red blood when urinating, blood clots, inability to urinate, fever >38C and chills/abdo pain unrelieved by analgesia.
- Tub/sitz bath to relieve discomfort/pain (except with recurrent urinary tract infections)

Meningitis

- Inflammation of the meninges covering the brain and spinal cord.
- Key s/s: fever, severe headache, N,V and nuchal rigidity.
- Other s/s: photophobia, decreased LOC, and ICP
- Diagnosis done through: lumbar puncture which assesses the CSF.
- As the client is hypotensive and septic, they first need to be given treatment in the following order:
 - 1) initiate fluids
 - 2) blood cultures taken for antibiotics
 - 3) a/bs given stat
 - 4) head CT scan after the client's BP is normal and treatment with fluids/vasopressors is normal to rule out ICP (altered mental status, seizures, neuro effects) and brain herniation risk.
 - 5) Lumbar puncture

West Nile Virus

- Mosquito borne disease (encephalitis) that occurs mainly during the summer months, especially during humid weather.
- Use insect repellent and prevention methods (long sleeves, long pants, light colours, avoid outdoor activities at dawn and dusk when mosquitoes are most active).
- Transmitted through an infected mosquito bite

Hep A and typhoid

- Transmitted fecal, oral and through contaminated food, water

Ringworm

- Superficial fungal(tinea) skin infection that mostly affects scalp or feet (athlete's foot)
- Limit contact with infected pets
- Spread through contact and indirect contact
- Fungus thrives in warm, moist areas.
- s/s: itchy, red, raised, scaly patches that may blister or ooze. Sharply defined edges. Red on the outside normal skin on the inside, looking like a ring.
- Tests: KOH exam, skin biopsy, skin culture, skin observation
- Keep skin clean, dry. Apply antifungals (miconazole, ketoconazole etc); shampoo regularly, especially after haircuts, do not share personal items, wear sandals and shoes at gyms, lockers, pools, avoid touching pets with bald spots.

Allergies/asthma (from mites) or scabies (contagious skin infection by mites)

- Wash bedding in hot water

Urine testing

- Clear catch or midstream urine samples are collected for urinalysis or urine culture and sensitivity testing.
- Creatinine clearance test: all urine for 24 hours must be collected. The first urine specimen is discarded in a container and kept cool and the time is noted. After 24 hours, the client should void one last time and add the specimen to the container. Blood is also drawn to measure the creatinine level. **Creatinine clearance is the measure of the glomerular function and is a sensitive indicator of renal disease progression.**
- An in and out catheter (straight catheter) is used for any rest requiring a urine specimen when the client is unable to urinate or unable to follow the specimen collection procedure. A catheter is also used for a cystourethrogram or a residual urine test.
- **The first AM void is preferable for a urinalysis or urine culture and sensitivity as an overnight specimen is more concentrated.**

Infected endocarditis:

- Check temperature regularly as persistent elevations could mean the a/b therapy is

ineffective or complications have developed. Client should notify HCP if fever persists at home.

- Client has risk of reoccurrence. They should receive prophylactic a/b before high risk procedures. Vegetations on valves and surfaces can form, and embolization to various organ sites can occur. Slurred speech could indicate that this has occurred, and could lead to a possible stroke. s/s: one sided weakness, slurred speech, paralysis, painful, cold extremities.
- a/bs for up to 4-6 weeks.

Morphine sulfate

- N, V are expected s/e of opioid meds when treatment is initiated. But, tolerance develops quickly and persistent. Take an anti-emetic with the pain med.
- N, V less likely to occur in a recumbent position, and risk increases by 40% in clients who are up and walking.
- Taking meds on empty stomach may increase risk of N
- N, V decreases when the pt lies in a flat position

Pain

- In the right lower quadrant: appendicitis. Pain starts in periumbilical region and migrates to the McBurney's point. Client will attempt to decrease pain by lying still, with right leg flexed and preventing intra abdominal pressure (avoid coughing, sneezing, deep inhalation)
- In the left lower quadrant: diverticulitis (often in the sigmoid colon). Other s/s include palpable tender abdominal mass, and systemic symptoms of infection (fever, increased C-reactive protein, and Leukocytosis with a left shift)
- C reactive protein: a by product of inflammation; a globulin that is found in the blood in some cases of acute inflammation. It's a protein made by the liver and released within a few hours of tissue injury, start of infection/cause of inflammation.
- Pain in the right upper quadrant referred to the right scapula: acute cholecystitis. Also experience indigestion, N, V, restlessness, and diaphoresis

Small Bowel Follow Through (SBFT)

- Examines the anatomy and function of the small intestine using X-ray images taken in succession.
- Barium is ingested and x-ray images are taken every 15-60 minutes to visualize the barium as it passes through the small intestine.
- This can help identify decreased motility, increased motility, fistulas, and obstructions.
- Clients should follow these instructions:
fast 8 hours prior to the exam. **Polyethylene glycol is prescribed as bowel prep for a colonoscopy NOT an SBFT.**

the test usually takes 60-120 mins, but it can take longer if obstruction or decreased mot.

Occurs

drink plenty of fluids after the exam to help remove barium. Chalky stools may be present for 24-72 hours after the exam. If brown stools don't return after 72 hours or abdo pain or fullness is present, contact HCP

- Black tarry stools are not expected, it can indicate a GI bleed. Report stat to HCP
- Endoscope is not used for this test

Celiac disease

- a disease in which the small intestine is hypersensitive to gluten, leading to difficulty in digesting food. This damage makes it hard for the body to absorb nutrients, especially fat, calcium, iron, and folate.
- Gluten is a form of protein found in some grains and should be eliminated from diet. This reduces the risk for nutritional deficiencies, and intestinal cancer (lymphoma).
- s/s: abdo bloating, pain, gas, diarrhea, pale stools, weight loss; skin rash, iron deficiency anemia; muscle cramps, joint and bone pain. Growth problems and failure to thrive. Seizures, tingling sensation (caused by nerve damage and low calcium); sores in mouth; missed menstrual periods
- gluten foods: wheat, rye, barley, oats.
- Processed foods (chocolate, candy, hot dogs) may contain hidden sources of gluten such as modified food starch, malt, and soy sauce. Food labels should indicate that the product is gluten free.
- Rice, corn, and potatoes are allowed.

Salem Sump Tube

- Checking for residual volume is not appropriate as the sump is attached to continuous suction for decompression and is not being used to administer enteral feeding.
- The air vent (blue pigtail) must remain open as it provides continuous flow of atmospheric air through the drainage tube at its distal end (to prevent excessive suction force). If the gastric content refluxes, 10-20 ml of air can be infected into the vent. And the vent is kept above the level of the client's stomach to prevent reflux.
- Place in semi-fowler's position to keep the tube from lying against the stomach wall to help prevent reflux.
- Provide mouth care every 4 hours to help maintain moisture of oral mucosa and promote client comfort.
- Turn off suction briefly during auscultation as the suction sound can be mistaken for bowel sounds
- Inspect the drainage system for patency (tube kinks, blockage)

Hiatal hernia

- Occur d/t a weakened diaphragm and increased intraabdominal pressure. The interventions are similar to those for GERD and focus on decreasing intraabdominal

pressure.

- Diet mod: avoid high fat foods, and those that decrease lower esophageal sphincter pressure (chocolate, peppermint, tomatoes, caffeine). Eat small, frequent meals; decrease fluid intake during meals to prevent distention. Avoid meals close to bedtime and nocturnal eating.
- Lifestyle changes: smoking cessation, weight loss
- Avoid lifting/straining
- Elevate the HOB to 30 degrees: this can be done at home using pillows or 4-6 in blocks under the bed.
- Wearing a girdle or tight clothes increases pressure and should be avoided.

Colostomy care:

- Ensure sufficient fluid intake (**at least 3,000 ml/day** unless contraindicated) to prevent dehydration. Identify times to increase fluid requirements (hot weather, increased perspiration, diarrhea)
- Identify and eliminate foods that cause gas and odor (broccoli, cauliflower, dried beans, Brussels sprouts)
- **Empty pouch when it becomes 1/3 full** to prevent leaks due to increasing pouch weight

PPIs (proton...)

- Are associated with decreased bone density (**calcium malabsorption**) which increases the possibility of fractures of the spine, hip and wrist.
- It causes acid suppression that otherwise would have prevented pathogens from more easily colonizing the upper GI tract. This leads to increased **risk of pneumonias**.
- It can also increase the risk of **C-diff associated diarrhea**. Unclear cause. Receiving a/bs for a UTI will further increase risk of C diff infection.
- Take medication prior to meals.
- PPIs don't affect BP
- Increase calcium and vitamin D intake to prevent osteoporosis.

Phenytoin

- Anticonvulsant
- Toxicity: gait disturbances (ataxia)

Bisphosphonate (Alendronate, risedronate)

- Bisphosphonates are a class of drugs that prevent the loss of bone mass, used to treat osteoporosis and similar diseases
- Drink extra water and stay upright for 30 mins after taking the drug to prevent

esophagitis.

- Jaw necrosis is a toxicity s/s

Lithium (mood stabilizing drug) and Albuterol (bronchodilator, short acting beta-agonist)

- Toxicity: tremors

Dialysis

- Peritonitis is the most common complication of dialysis.
- Chills may indicate elevated temp (sign of infection) and rebound tenderness a sign of peritonitis (inflammation of the peritoneal cavity, with cloudy effluent). Abdo pain also present.
- Assess peritoneal fluid for C&S.
- Clients receiving peritoneal dialysis often have diabetes, and glucose (dextrose) is the osmotic agent in dialysate. Monitor glucose levels closely. Regular insulin can be added to the dialysate before the solution is instilled or it can be administered subQ to control glucose levels.

Ulcerative colitis

- Chronic disease , inflammation of the large intestine
- Results in urgent, frequent bloody diarrhea; abdo pain, anorexia; and anemia.

The gFOBT (guaiac fecal occult blood test)

- Correct sequence of test:
Obtain supplies, wash hands, don non-sterile gloves
Open the slide's flap and use the wooden applicator to apply 2 separate stool samples to the boxes on the side
Wait 3-5 mins
Open the back of the slide and apply 2 drops of developing solution to the boxes on the side
Wait 30-60 seconds
Document the results in the EMR
- This test is used to assess the microscopic blood in the stool and as a screening tool for colorectal cancer.
- Assess recent consumption of red meat or Vit C in the last 3 days, or using certain meds (aspirin, anticoagulants, iron, ibuprofen, and corticosteroids) as they can interfere with the results.
- If the test paper **turns blue, the test is positive** and the stool contains blood.

Exacerbations tested for HF

- BNP: B-type natriuretic peptides are made, stored, and released primarily by the ventricles. They are produced in response to stretching of the ventricles d/t increased blood volume and higher levels of extracellular fluid (fluid overload) that accompany with HF.
- Elevation of BNP >100 pg/mL helps to distinguish cardiac from resp causes of dyspnea.

MAP

- Average pressure within the arterial system felt by the vital organs. A MAP > 60 is necessary to adequately perfuse and sustain the organs. When it falls below this number, the organs can be underperfused and become ischemic.
- Normal MAP: 60-105 mm Hg.
- Formula:
$$SBP + (DBP \times 2) / 3$$

Systemic Lupus

- An autoimmune disorder in which an abnormal immune response leads to chronic inflammation of different parts of the body.
- Mild (skin, muscles, joints)
- Severe (affecting kidneys, heart, lung, blood vessels, CNS)
- Lupus nephritis: increased creatinine, BUN, and abnormal urinalysis (protein RBCs, cellular casts).
- A positive ANA (antinuclear antibody titer, >1:40); elevated ESR; anemia, mild leukopenia (<4,000) and thrombocytopenia (<150,000) indicates the presence of SLE and is expected.

Cardiac Tamponade

- Complication of pericardial effusion
- s/s: muffled or distant heart tones, narrowed pulse pressure, jugular vein distention, pulse paradoxus (widening systolic pressure of >10), dyspnea, tachypnea, tachycardia.
- Treatment: emergency pericardiocentesis
- Beck's triad: hypotension, muffled heart sounds, distended neck veins

CSF

- Assessed for color, contents, and pressure.
- Normal CSF is clear and colorless, and contains little protein, glucose, minimal WBCs, **NO RBCs**, and no microorganisms.

- Normal CSF pressure is 60-150 mmH2O.
- CSF is collected via lumbar puncture or ventriculostomy.
- BEFORE a lumbar puncture:
Empty the bladder before procedure
Performed in the lateral recumbent or sitting upright position. Help widen space between the vertebrae and allow easier insertion of the needle.
A sterile needle will be inserted between the L3/4 or L4/5 interspace.
Pain may be felt radiating down the leg, but it should be temporary.

AFTER the procedure:

Lie flat with no pillow for at least 6 hours to reduce the chance of spinal fluid leak and resultant headache.

Increase fluid intake for at least 24 hours to prevent dehydration

Bell's palsy

- Inflammation of cranial nerve VII → motor / sensory alterations
- Corticosteroids to reduce inflammation, and taught eye/oral care.
- Eyelids don't close properly, causing eye dryness and risk of corneal abrasions.
- Weakness of the lower eyelid may cause excessive tearing due to overflow in some clients.
- Facial muscle weakness results in poor chewing and food retention.
- Pt. teaching:
Eye care: use glasses during the day; wear a patch or tape the eyelids at night to protect the exposed eye. Use artificial tears during the day as needed to prevent excess drying of the cornea.
Oral care: chew on unaffected side, to prevent food trapping; a soft diet is recommended. Maintain good oral hygiene after every meal to prevent problems from accumulated food (parotitis, dental caries)
- Vision, balance, consciousness, and extremity motor function are not impaired
- Assessment:

Ask pt. to smile, frown, raise eyebrows, open lips to show teeth to assess this nerve.

Any asymmetrical movements are documented, and if unexpected, the HCP can be notified.

MI

- BP and serum K levels are checked prior to administration of ACE inhibitors.
- HR checked prior to beta blockers
- Aspirin is given to clients with normal platelet counts (150,000-200,000)

- Simvastatin is a lipid lowering med given to all clients with elevated triglycerides and LDL cholesterol. It is usually given in the evening.
- Metoprolol given post MI to reduce the risk of re-infarction and occurrence of HF. s/e: Bradycardia. Hold med and notify HCP.
- Docusate sodium, stool softener. Clients should not strain during a bowel movement d/t risk of producing a vagal response, putting the client at risk for Bradycardia, and other dysrhythmias.
- Aspirin given as antiplatelet to help reduce inflammation and inhibit platelet aggregation. Assess platelet count, hemoglobin and hematocrit count prior to administration.
- Lisinopril, ACE inhibitor, given post MI to prevent ventricular remodeling (hypertrophy), and progression of HF. They can cause hyperkalemia, so need to assess K level. Assess BP prior to administration, as it can lower BP.

Troponin

- Cardiac specific serum marker that is a highly specific indicator of MI and has greater sensitivity and specificity for MI injury than CK or CK-MB,
- Serum troponin I increase 4-6 H after onset of MI, peak at 10-24 H and return to baseline in 10-14 days.
- However, MI is not diagnosed alone by serum cardiac markers. ECG and client health history along with history of pain and risk factors are also used to make diagnosis.

Antihypertensives

- Can cause rebound hypertension and a possible hypertensive crisis if not taking medications. This is a major issue in long-term management of HTN (poor adherence).
- s/e: fatigue, dizziness, erectile dysfunction.

Nicardipine (cardene) a prototype of Nifedipine given during a stroke presentation (brain attack)

- CCB vasodilator and takes effect 1 min of IV administration.

- Monitor BP and ensure it does not fall too quickly or slowly as this would extend a stroke. Hypotension can occur, with or w/o reflex tachycardia. d/c drug if this happens.
- Permissive HTN during first 24-48H allows for adequate perfusion through damaged cerebral tissues. The blood brain barrier is not longer intact after the BP reaches >220/120. Need to lower the BP gradually, and the SBP should not fall below 170.

Torsades de pointes

- Ventricular tachycardia, due to increased magnesium
- Occurs with haloperidol, methadone (analgesic), ziprasidone (antipsychotic), erythromycin (antibiotic to treat gram positive infections, similar in effects to PCN)

Spironolactone (Aldactone)

- K sparing diuretic

BNP

- Hormone released by the heart muscle in response to mechanical stress (stretching). BNP levels are usually elevated (>100 pg/ml) in clients with HF, and the prescription of Lasix is expected.

Atrial Fib

- Total disorganization of atrial electrical activity resulting in the loss of effective atrial activity resulting in the loss of effective atrial contraction.
- Atrial rate can be 350-600/min.
- P waves are not visible. They are replaced by fibrillatory waves. Ventricular rate varies, but the rhythm is typically irregular.
- Results in: decreased CO, d/t loss of atrial kick and/or a rapid ventricular response. Clots may form in the atria, putting the client at an increased risk of stroke.
- Treatment: rate control, and anticoagulation.

Atrial Flutter

- Recurring, regular, sawtoothed-shaped flutter waves.
- Rate: 200-350

Complete Heart Block

- The presence of more P waves than QRS complexes. The PR interval is variable. There is no communication between the atria and ventricles. Each is firing independently at each other.

MVP (mitral valve prolapse)

- may have palpitations, dizziness, and light-headedness. Chest pain can occur, but its etiology is unknown in this client population.
- **Chest pain does not typically respond to anticoagulant treatment such as nitrates.** Beta blockers may be prescribed for palpitations and chest pain.
- Interventions:
 Adopt healthy eating habits and avoid caffeine as it stimulates and may exacerbate the symptoms
 Check OTC meds or diet pills for stimulants such as caffeine or ephedrine as they can exacerbate symptoms.
 Reduce stress, and avoid alcohol use.
 Begin an exercise program, preferably aerobic exercise, to achieve optimal health.
 A/bs is indicated for clients who have prosthetic valve replacement, repaired valves, or a history of infectious endocarditis, as MVPs may place the client at an increased risk for infective endocarditis. Prophylactic a/b before dental procedures are not indicated.
No need for a medical alert bracelet, as MVPs are usually a benign condition.

beta blocker

- any of various drugs used in treating hypertension or arrhythmia; decreases force and rate of heart contractions by blocking beta-adrenergic receptors of the autonomic nervous system

Coronary Arteriogram (angiogram)

- Invasive diagnostic study of the coronary arteries, heart chambers, and function of the heart.
- Client should have an IV line started for sedating meds. The femoral or radial artery will be accessed during the procedure. General anesthesia is not given.
- Instruct the clients:
 Do not eat or drink anything 6-12 hours prior to the procedure (depending on the particular health care provider performing the procedure)
 Client may feel warm or flushed while the contrast dye is being injected
 Homeostasis must be obtained in the artery that was cannulated for the procedure. Mostly the femoral artery. Compression is applied to the puncture site and the client may have to lie flat for several hours to ensure homeostasis.
- If the procedure is just a diagnostic study, the client will often go home the same day.

Hospitalization for 1-3 days may be required if angioplasty or stent replacement is performed.

Automatic dysreflexia (hyperreflexia)

- Massive, uncompensated cardiovascular reaction by the SNS in spinal injury at T6 or higher. Due to an injury, the PNS cannot counteract the SNS stimulation below the injury. Classic triggers are distended bladder or rectum d/t bladder irritation, bowel impaction.
- S/s: severe HTN (up to 300 mm Hg systolic) , throbbing headache, diaphoresis above the injury level, Bradycardia (30-40/min) , piloerection (goose bumps), nausea and flushing. This is an emergency situation requiring immediate intervention to prevent hypertensive stroke and seizures.
- Management:
Raise HOB and then treat the cause.
Catheter the pt if needed, and assess for kinks in existing catheters
Perform digital rectal exam.
Remove constrictive clothing to decrease skin stimulation.
Contact HCP
Alpha adrenergic blocker, or an arteriolar vasodilator (Nifedipine) may be prescribed.
Elevate HOV to 45 degrees or High fowlers to decrease HTN. The Sim's position is supine and side-lying and not the right position.

Myasthenia gravis

- Autoimmune disease→ muscle weakness and ptosis. Weakness increases with activity, and ptosis is present at the end of the day. These are expected findings. Respiratory failure is possible, so monitor closely.

Transverse myelitis (spinal cord inflammation)

- Usually results from a recent viral infection. classic symptoms include paralysis, urinary retention, and bowel incontinence. Some recover, others have permanent disability.
- Normal muscle strength: 5 on a 0-5 scale. Weakened muscle strength (2+ means only able to move laterally, not able to lift up against gravity.. this is expected.

PSVT (paroxysmal supraventricular tachycardia)

- HR: 150-220/min
- Prolonged episodes can = hypotension, palpitations, dyspnea, angina
- Treatment:

Vagal maneuvers (Valsalva, coughing, carotid massage)

Adenosine is the DOC, administered rapidly via IVP over 1-2 seconds and followed by 20 ml saline bolus d/t its short half-life. An increased dose can be given twice if the previous admin. is ineffective.

(Adenosine is an antiarrhythmic and a nucleoside. It works to treat irregular heartbeat by slowing the electrical conduction in the heart, slowing heart rate, or normalizing heart rhythm. It helps during a stress test of the heart by improving blood flow to the heart.)

Beta-blockers, CCBs, and amiodarone, can be considered as alternatives.

If drug therapy is useless, synchronized cardioversion may be used.

Defibrillation

- Used only in clients with V fibrillation and pulseless V tach.

External pacing and atropine (anticholinergic used to increase HR)

- Used in symptomatic bradycardia (<60/min)

Stroke

- Single most important modifiable risk factor: HTN
- Other risks: diabetes, high cholesterol, smoking, obesity (particularly abdominal), older age, and genetic susceptibility.
- Acute stroke treatment: can give Labetol (Normodyne) or nicardipine (Cardene) as antihypertensives.
- Permissive HTN is allowed within the first 24-48 hours of an acute ischemic stroke provided the BP is <220/110 mm Hg. This allows the penumbra to be perfused to keep the stroke from extending.

Myasthenia gravis

- An autoimmune disease involving a decreased number of acetylcholine receptors at the neuromuscular junction. Results in fluctuating weakness of skeletal muscles, most often shown as ptosis/diplopia, bulbar signs (difficulty speaking or swallowing); and difficulty breathing.
- Muscles stronger in the morning, and become weaker with the day's activities as the supply of acetylcholine is depleted.
- Skeletal muscles, not reflex or CNS muscles (for bowel and bladder control) are involved in Myasthenia gravis. Bladder issues are related to MS.
- It affects eye and eyelid movements, speaking, swallowing, and breathing.

- Treatment:

Anticholinesterase drugs (mestinon/pyridostigmine) that are administered before meals so that the client's swallowing ability is strongest during the meal.

Semi-solid foods that are easy to chew are preferred to avoid stressing muscles involved in chewing and swallowing, or liquids (aspiration risk).

Annual flu vaccine (also pneumonia vac. If appropriate) as they are more likely to have a negative outcome if the illness is contracted.

Anticholinergic drug (atropine) is used to treat cholinergic crisis (too much acetylcholine); the need for this drug would not be needed during a myasthenic crisis, as it is a result of too little medication related to noncompliance, illness, or surgery.

Pulmonary Edema

- It's a life threatening condition.
- Clinical s/s:
History of orthopnea (dyspnea while lying down) and/or paroxysmal nocturnal dyspnea
Anxiety/restlessness
Tachypnea (often >30/min), dyspnea, and use of accessory muscles
Frothy, blood-tinged sputum
Crackles on auscultation
- Priority care: improve oxygenation by reducing pulmonary pressure and congestion.
Diuretics (Lasix) are prescribed to remove the excess fluid in pulmonary edema.

Theophylline (theo-24)

- A long-acting, slow release bronchodilator that relaxes bronchial smooth muscles that improves contractility of the diaphragm, and helps muscle transport by the cilia.
- It has a narrow therapeutic index, and toxicity can occur from accumulation by reduced clearance or decreased metabolism.
- Monitor serum drug levels every 6 months to maintain a target level of 10-20 mcg/mL; symptom management can be attained at a lower target range (8-15 mcg/mL)

SIRS

- occurs during a major insult (trauma, infection, burns, bleeding, transfusions). Leads to stimulation of immune response, increased WBCs, increased cap permeability, and inflammation of organs.

Sepsis→severe sepsis→septic shock→MODS (multiple organ dysfunction_

Diagnostic criteria:

Hyperthermia ($>100.4/38$) or hypothermia ($<97/36.1$)

HR: $>90/\text{min}$

RR: $>20/\text{min}$ or alkalosis ($\text{paCO}_2 <32\text{mmHg}$)

Leukocytosis (WBCs $>12,000$ or 10% immature neutrophils –bands)

CVP: normal : 2-6 mmHg

-indicates circulating volume. It is decreased not increased, in septic shock d/t massive vasodilation and misdistribution of blood flow.

MAP: normal: 70-100 mmHg, and is not associated with SIRS.

Carbon Monoxide Poisoning

- pulse oximeter reading is normal because it cannot differentiate hemoglobin saturation with CO versus O₂.
- Nonspecific s/s: headache, dizzy, fatigue, N, dyspnea)
- Serum carboxyhemoglobin test is needed to confirm the diagnosis. Normal values are 2-3% in non-smokers and slightly higher in smokers.
- Requires immediate admin of 100% O₂ to correct hypoxemia and eliminate CO from blood.

Methylphenidate (Ritalin, Concerta)

- CNS stimulant used to treat ADHD and narcolepsy. It contributes to hyperactivity and lack of impulse control by affecting neurotransmitters (dopamine, norepinephrine)
- Common side effect: loss of appetite, resulting in weight loss.
- Parents and caregivers should weigh child with ADHD at least weekly d/t the risk of temporary interruption of growth and development. Compare weight/height measures from one well child checkup to the next.
- Increased BP and tachycardia can also be a side effect. Monitor before and after starting treatments.

Statins

- Significant s/e: muscle aches (myopathy)/weakness.
- Call HCP who will then obtain a blood sample to assess the CK levels. If the CK is significantly elevated ($\geq 10\times\text{normal}$) the drug will be d/c.

Benzodiazepines

- Antianxiety drugs.
- May cause sedation, which can interfere with daytime activities. Giving dose at bedtime will help the client sleep.
- Never stop abruptly. It should be tapered gradually to prevent rebound anxiety and a withdrawal reaction (anxiety, confusion, etc).

Monoamine oxidase inhibitors (Tranylcypromine, phenelzine)

- Used for depressive disorders.
- Eliminate cheese, processed meats, and other tyramine containing items.
- Could cause photosensitivity.

Aspirin toxicity

- Tinnitus

Hemophilia

- Bleeding disorder d/t deficiency in coagulation proteins. Clients with classic hemophilia or hemophilia A lack factor VIII. Those with Hemo. B (Christmas disease) lack factor IX.
- Most frequent sites of bleeding are the joints, especially the knee. Hemarthrosis can occur with minimal or no trauma, with episodes beginning during toddlerhood when the child is active and ambulatory.
- Over time, chronic swelling and deformity can occur.

Methotrexate

- Used in treating rheumatoid arthritis and psoriasis
- Folate antimetabolite, Antineoplastic, immunosuppressant drug used to treat various malignancies as a nonbiologic disease-modifying antirheumatic drug (DMARD)
- Can cause bone marrow suppression resulting in anemia, leukopenia, and thrombocytopenia.
- Increased susceptibility to infection.
- Obtain vaccines, avoid crowds, and persons with known infections.
- Herpes zoster (live vaccine) is contraindicated. Inactivated vaccines (influenza and pneumococcal) are allowed
- Avoid alcohol as it is hepatotoxic.

- GI irritation
- Eyes do not need to be checked every 6 months for this drug. it does need to be checked for the DMARD antimalarial drug hydroxychloroquine (plaquenil) as it can cause retinal damage.
- **Can cause congenital anomalies and fetal death. Client should not become pregnant while taking this drug or for 3 months after it is d/c. It is teratogenic.**

Photosensitivity

- Tetracyclines, thiazide diuretics, sulfonamides

Nephrotoxicity

- Aminoglycosides, vancomycin, and NSAIDS

12 month old infant

- Equal head and chest circumference.
- Triple birth weight
- Sits from a standing position without help
- Prefers parents and exhibits fear when separated
- Should have about 6 teeth
- Should attempt to place small objects into a narrow opening but is unsuccessful

INH (Isoniazid)

- Used for TB
- It interferes with the action of Vitamin B6 resulting in peripheral neuropathy (ataxia and paresthesia)
- Those most prone to these are older adults, malnourished people, diabetic clients, pregnancy or breastfeeding clients, alcoholics, children, those with liver/renal disease, and HIV+ individuals.
- Give VitB6 at 25-50 mg/day for those at high risk.
- Neurological s/e can also occur.

Spironolactone (Aldactone)

- Potassium sparing diuretic that counteracts K loss by other diuretics (thiazides)
- Helps treat HTN, ascites (d/t liver disease)

Questran

- May be given to excrete bile salts in the feces to reduce pruritus (during cirrhosis).
- Powder form, must be mixed in foods (applesauce) or juice (apple juice) and should be given one hour after all other meds.

Malignant Hyperthermia

- hereditary condition in which certain anesthetics (e.g., halothane,) (andanecline, succinylcholine a muscle relaxant) cause high body temperatures and muscle rigidity
- do proper screening and a thorough pre-op nursing assessment/health history to minimize client risk.

Moderate to Severe Asthma

- tachycardia >120/min
- tachypnea >30/min
- saturation <90% on room air
- use of accessory muscles to breathe
- PEF <40% of predicted or best (<150L/min)
- Need to correct hypoxemia, improve ventilation, and promote bronchodilation by:
Maintaining O2 Sat >90%
High dose inhaled short-acting beta agonist (SABA) **Albuterol** and anticholinergic agent (**ipratropium**) nebulizer treatments every 20 mins
Systemic corticosteroids (**Solu-Medrol**)
O2 to maintain saturation >90

Agoraphobic fears:

- Being outside home alone
- Being in a crowd or standing in line
- Traveling in a bus, train, car, ship, airplane
- Being on a bridge or in a tunnel
- Being in open spaces (parking lots, market places)
- Being in enclosed spaces (theaters, concert halls, stores)

Its intense anxiety about being in a situation from which there may be difficulty escaping in the event of a panic attack.

Tinea corporis (ringworm)

- Fungal infection of the skin often transmitted from person to person, or infected animal to person.
- Scaly, pruritic patch, circular or oval.
- Highly contagious and can be spread via grooming tools, hats, towels, bedding.
- Often spreads by sharing athletic equipment in locker rooms due to close proximity of infected gear.
- Treat with topical antifungals (tolnaftate, haloprognin, miconazole, clotrimazole)

Ear Drops

>3 years old

- pull pinna of ear upward and back to straighten the external ear canal.

<3 years old

- Pull pinna down and straight back

The child should be placed in the prone/supine position with the head turned to the appropriate side

Otic meds should be warmed to room temp if removed from a refrigerator prior to administration. Holding the bottle in the palm of the hand is an effective method of warming.

Instilling cold drops in the ear can cause a vestibular reaction, resulting in dizziness, and vomiting.

Child should remain with the affected ear up for several minutes after administration to allow full coverage of the med.

Enteral feedings

- Preferred over TPN feedings, as it maintains the integrity of the gut, prevents stress ulcers, and helps prevent the translocation of bacteria into the bloodstream.
- Complications: aspiration, tube displacement, hyperglycemia, diarrhea, abdominal distention, enteral tube misconnections, and clogged tubes.

Failure to Thrive

- Weight less than 80% of ideal for age and or depressed weight for length.
- Underlying cause: inadequate dietary intake

- Contributing factor: disturbance in feeding behavior, and psychosocial factors.
- Observe the child first when feeding so nurse can identify potential factors contributing to insufficient intake.

Amoxicillin/clavulanate

- Treats respiratory infections.
- Instructions:
May be taken with or w/o food as it does not affect absorption
N,V, diarrhea are most common s/e. Take with food if these occur to decrease GI s/e
Shake the liquid well prior to administration. Give at evenly spaced intervals throughout the day to maintain therapeutic blood levels
Ensure full course of therapy is taken, do not d/c the med even when feeling better or symptoms are resolved.

Rash, itching, dyspnea, or facial/laryngeal edema may indicate an allergic reaction, and the medication should be d/c

Nocturnal enuresis

- Pharmacological interventions used as second line treatment for children age >5 years.
- This is done when there has been no response to behavioral approaches and/or when short-term improvement of enuresis is desired for attending sleepovers/overnight camp.
- Trial run usually done at least 6 weeks before camp to see if the drug is appropriate and effective. There is a high risk of prolapse however, once the drug is d/c.
- Medications used to treat this:
Desmopressin, reduces the urine production during sleep.
Tricyclic antidepressants (imipramine, Amitriptyline, desipramine) which help improve the functional bladder capacity.

Intussusception

- Is an intestinal obstruction that occurs when a segment of a bowel folds (telescopes) into another segment.
- It causes increased pressure, causing ischemia, and leakage of blood and mucous into the lumen.
- Stool mixed with mucus (red currant jelly).
- Initially, infants may only have general symptoms (irritability, diarrhea, lethargy).
- Subsequently (sudden, abdominal pain(cramping), drawing the knees up to the chest, and inconsolable crying, are seen. After an episode, the infant may vomit and then appear otherwise normal.
- Assessment may show sausage shaped abdominal mass.

Oily, bulky, foul-smelling stools

- Excess fat in stool (steatorrhea) from malabsorption.
- Characteristic of pancreatic insufficiency, cystic fibrosis, or celiac disease.

Dark red, black sticky stools

- Upper GI bleed, gastritis (in infants and toddlers)

Hirschsprung disease (congenital aganglionic megacolon)

- Thin, ribbon like stools. Bowel obstruction is d/t failure of the internal sphincter to relax.

Beta blockers (“lols”)

- **Used for CHRONIC heart failure**
- In acute decompensated HF (ADHF) these drugs can further deteriorate the client's condition. (marginally low BP, crackles in lungs, low O2 sat, jugular vein distention, peripheral edema).

Statins

- Contraindicated in severe liver or muscle injury.

Anaphylactic Shock

Call for help-first action

Maintain airway, breathing-administer high flow O2 via nonrebreather mask

Epinephrine, intramuscular-DOC. Dilates and constricts. If no response, repeat Q5-15mins

Elevate legs

Volume resuscitation with IV fluids

Bronchodilator (Albuterol) to dilate small airways and reverse constriction

Antihistamine (diphenhydramine) to modify hypersensitivity reaction and relieve pruritus

Corticosteroids (Solu-medrol-methylprednisolone) to decrease airway inflammation and swelling d/t allergic reaction.

Creatinine level: 0.6-1.3 mg/dL

Ace inhibitors: hyperkalemia and hypotension are contraindications for this drug. these drugs are excreted renally and can worsen renal function, so evaluation of the kidney function is essential for clients taking these meds. (same for Aminoglycosides-gentamicin) and digoxin

Normal fetal heart tones: 110-160/min

Normal pulse rate in a 5 year old: 70-120/min average is 100

Albuterol(Ventolin) is a bronchodilator beta-adrenergic, and the expected s/e are: tremors, tachycardia, palpitations

DONNING

Hand hygiene

Gown

Mask/respirator

Goggles/face shield

Gloves

Hypokalemia: muscle cramps in legs, weakness, paralysis

Epiglottitis

- Haemophilus influenza type B vaccine reduces the incidence due.
- It's an inflammation by bacteria of the tissues surrounding the epiglottis. Edema can develop and obstruct the airway, occluding the trachea.
- Classic symptoms: high grade fever with toxic appearance, severe sore throat, and the 4 D's: dysphonia (muffled voice), dysphagia (difficulty swallowing), drooling, and distressed respiratory effort.
- Place pt. in tripod position. It helps open the airflow. Sit rather than lie down.

ICP in a child

- Wide, bulging fontanelle
- Prominent scalp veins
- Increased head circumference
- Sunset eyes (6th cranial nerve palsy (paralysis of upward gaze) as a result of ICP, hydrocephalus. It's an acute, delayed sign and requires timely priority diagnosis and

treatment. This sign is more likely to be noted after the fontanelles have closed (posterior by 6 and anterior by 18 months) and the pressure increases.

-

Metoclopramide

- Antiemetic and/or prokinetic agent that promotes GI motility and gastric emptying.
- Commonly used to treat N, V and gastroparesis.
- It can cause tardive dyskinesia (TD) a condition characterized by unusual uncontrollable movements of the arms, legs, head, and face, or entire body. (twisting/protruding tongue movements, lip smacking, torticollis, and piano-playing finger movements)
- TD is irreversible in many cases, and the risk for developing this kind of TD is greater with advanced age, long term therapy, and high drug doses.
- Diarrhea is not a serious s/e. Burping does not occur as TD does not affect HR.

Chest drainage

- >100 ml/hr should be reported to the HCP. Large losses of blood may indicate a compromise of the surgical suture site and may require repair. Patient can become hemodynamically unstable.

Snellen Letter chart

- Tests visual acuity in those ages 6 and older
- Position 10 ft (3 m) from chart and ask to read letters.
- Standard testing for visual acuity is 20 ft (6m) but it's easier to maintain child's attention with 10 and provides more accurate results.
- If child wears glasses, they remain in place.
- Child must identify 4 of 6 letters in each line before moving to the next.
- Refer to an ophthalmologist if child is unable to identify 4 correct letters on the 10/15 line (equivalent to 20/30 vision) with either eye.

Following a bright colored object or a human face

- Method to check acuity and fixation in infants. If this is not present by 3-4 months, referral is made for a formal ophthalmic exam.

Fat embolism

- Life threatening and occurs in fractures, especially those of the long bone and pelvis.
- Globules of fat leave the bone and travel through the bloodstream to the lungs, skin, and

brain where they can occlude the small vessels.

- Altered mental status will result from blocked blood vessels in the brain.
- Embolism to the lung would result in respiratory distress.
- Hallmark sign of fat emboli=presence of petechiae (pin sized red/purple spots) that result from small-vessel clotting and appear across the chest, in the axillae and in the soft palate.
- Respiratory distress, mental status changes, and petechiae are the classic manifestations.

Asthma

- NSAIDS(ibuprofen-motrin; and aspirin)and Beta blockers have the potential to cause problems for clients with asthma. 10-20% of asthmatics are sensitive to these meds and can experience severe bronchospasms after ingestion. This is prevalent in clients with nasal polyposis
- Guaifenesin(mucinex)- an expectorant used to facilitate mobilization of mucus and should not have the potential to exacerbate asthma or cause an attack.
- Loratadine (Claritin)- an antihistamine
- Vit D: helps maintain bone density

Addisonian crisis

- Addison's disease is adrenocortical insufficiency or hypofunction of the adrenal cortex.
- Addisonian crisis or acute adrenal insufficiency is a potentially life-threatening complication.
- N, V, abdo pain, hypotension, tachycardia, dehydration, hyperkalemia, hyponatremia, hypoglycemia, fever, weakness, confusion.
- Priority emergency management: shock treatment, fluid resuscitation (0.9 saline and 5% dextrose) and administration of high dose hydrocortisone replacement IV push

Pleural Effusion

- Abnormal collection of fluid (>15-20 ml) in the pleural space between the parietal and visceral pleurae that prevents lungs from expanding fully
- Causes decreased lung volume, Atelectasis, and ineffective gas exchange
- Clients commonly have dyspnea on exertion, and non-productive cough, diminished breath sounds, dullness to percussion, and decreased tactile fremitus.
- If the effusion is large, the trachea (mediastinum) is deviated to the opposite side
- Dullness with pleural effusions and pneumonia
- Hyperresonance with Pneumothorax (percussion shows hyperresonance)
- Wheezing shows obstructive process (asthma, COPD) not in pleural effusion

Kawasaki Disease

- a disease of unknown cause, occurring primarily in young children and giving rise to a rash, glandular swelling, and sometimes damage to the heart.
-
- Treated with aspirin and IVIG to prevent coronary artery aneurysms. Antibodies acquired from the IVIG (immunoglobulin) therapy will remain in the body for up to 11 months and may interfere with the desired immune response. To live vaccines. Therefore, live vaccines (varicella, MMR) should be delayed for 11 months after IVIG administration as this therapy may decrease the child's ability to produce the appropriate amount of antibodies to provide lifelong immunity.

Haemophilus influenza type b (Hib) and PCV (pneumococcal conjugate)

- Not a live vaccine, and final dose (4th) is recommended between ages 12-15 months, according to the CDC.

Hep B

- Not a live vaccine, the 3rd (final) dose should be given between ages 6-18 months.

Order of Head to Toe Assessment for Infants

- Auscultate first, as the infant is quiet. This allows the nurse to clearly hear the heart and lung sounds, and efficiently count the heart rates and respirations.
- Palpation and percussion is next, and are performed together in a head to toe i=direction, when the infant is still calm. This helps the nurse to assess while the abdo muscles are still relaxed.
- Traumatic procedures are performed towards the end (eyes, ears, mouth while crying).
- Reflexes (grasping, babinski) are performed last, and the last step is doing the Moro reflex, as the infant is moving around and awake at this point. The expected response to a sudden dropping or jarring motion is a reflexive startle and crying.
- Auscultate heart and lungs, palpate fontanelles, Percuss abdomen, inspect eyes, elicit moro reflex

Allopurinol

- To prevent gout attacks (pain and inflammation in joints caused by uric acid deposits)
- It inhibits uric acid production and improves solubility
- Gout is a build up of uric acid deposits in the joints that cause pain and inflammation.

This med helps to prevent uric acid deposits in the joints and the formation of uric acid kidney stones

- Take with a full glass of water, and educate client about good fluid intake as it will help prevent the formation of renal stones and promote diuresis (increase drug and uric acid excretion). This is the priority nursing action.
- Biosynthesis of uric acid occurs in the liver, and anti gout meds are excreted via the kidneys, so need to assess liver and renal fxn tests. Blood counts need to be monitored as well, as some anti gout meds can cause blood dyscrasias (abnormal state of the fluid).

Any rash, even if mild, should be reported ASAP to the HCP, and stop taking the med immediately, schedule an appointment, and notify the HCP. Even more serious hypersensitivity reactions can follow this: stevens-johnson syndrome and toxic epidermal necrolysis (life threatening issue that causes the epidermis to separate from the dermis).

- This drug takes several months to become effective, and is used to treating gout attacks. It's not effective in treating acute attacks. Clients will need to continue to take anti-inflammatories (NSAIDs and colchicines) for acute attacks.

Metoprolol, bisoprolol, carvedilol (beta blockers)

- Used for chronic heart failure

ACE inhibitors (prils) and angiotensin II receptor blockers (ARBs) (sartans)

- Create a risk for hyperkalemia.
- Ace inhibitors decrease the excretion of aldosterone, increasing K.

Cirrhosis and Lactulose

- Lactulose helps excrete ammonia in cirrhosis with hepatic encephalopathy. Achieve 2-3 soft stools/day.

Vancomycin

- Should be infused over at least 60 mins (100 mins if infusing ≥ 1 gram).

Isosorbide and hydralazine are used in African Americans with HF

- It reduces preload and after load, decreasing cardiac workload

Gestational hypertension

- $\geq 140/90$ that occurs after 20 weeks gestation without proteinuria
- Preeclampsia occurs when proteinuria or signs of end organ dysfunction occurs.

Pregnancy

- Fetal tachy: >160 beats/min for >10 mins. The most sensitive indicators of fetus health is fetus movement and fetal HR
- 4 movements/Hour or 10 movements in 2 hours is a reassuring finding
- Braxton-hicks contractions are felt mid pregnancy onward. These painless, occasional physiological contractions are normal. They are a concern if they become regular and persistent.

Metabolic syndrome

- Factors that increase the risk of CVDs and diabetes mellitus
- Criteria for metabolic syndrome include the presence of 3 or more of these following conditions:
 - 1) abnormal obesity with increased waist circumference (≥ 40 inches in men, and ≥ 35 in women)
 - 2) hyperglycemia ≥ 110
 - 3) low HDL (<40 mg/dL in men; <50 in women)
 - 4) high triglycerides ≥ 150 mg/dL
 - 5) hypertension: $\geq 130/85$

Lithium

- Mood stabilizer used to treat bipolar affective disorders.
- Therapeutic index: 0.6-1.2. levels >1.5 are considered toxic.
- Toxicity:
 - Dehydration
 - Decreased renal function (elderly)
 - Diet low in Na
 - Drug drug interactions (NSAIDs and thiazide diuretics)
- Lithium is cleared renally, even mild changes in kidney function can cause serious lithium toxicity. Drugs that decrease renal blood flow (NSAIDs) should be avoided.
- Acetaminophen would be a better choice for pain relief.
- Blood should be drawn frequently to monitor for therapeutic levels and toxicity.

FIRE ALARM

- RACE
Rescue clients from immediate danger and move to safety
Alarm and activate the agency's fire response
Confine the fire by closing the doors to all rooms and fire doors to the entrance of the unit
Extinguish the fire if possible with the fire extinguisher

Liver biopsy

- Position on right side afterwards for a minimum of 2-4 hours to splint the incision site. Client stays on bed rest for 12-14 hours.
- Lie supine with right arm over head with client holding breath before needle is inserted between the 6 and 7 or 8 and 9 ribs.

Sickle Cell crisis (vaso-occlusive crisis)

- Elevated bilirubin levels d/t the breakdown of hemoglobin and excess hemolysis. When this is 2-3X the normal level, jaundice results.
 - Elevated reticulocyte count
 - Hemoglobin <10 g/dL, resulting in anemia
-

28 weeks baby

- Newborn should have some adipose tissue, the ability to perform gas exchange if needed, and open eyes. This knowledge assists the nurse in assessing the gestational age of the newborn.
- Bones not fully developed until 32 weeks gestation
- Lanugo begins to disappear at 36 weeks gestation
- The lungs and respiratory system are not fully developed until 40 weeks gestation.

Hemophilia

- Seen primarily in males and is due to a lack of clotting factors.
- Symptoms include spontaneous bleeding (hemarthrosis) into the joints, (knees, ankle, elbow)
- Treatment: replace the missing clotting factor. Desmopressin (DDAVP) stimulates the release of factor VIII (blood clotting protein)
- Desmopressin is a vasopressin that helps retain fluid and constrict blood vessels

Lab values

Hct: 39-50%

Hbg: 13.2-17.3 g/dL

Thiazide diuretics

- Help treat HTN
- Hydrochlorothiazide and chlorthalidone
- Major side effects:
Hypokalemia (shown as muscle cramps)
Hyponatremia (altered mental status and seizures)
Hyperuricemia (may worsen gout attacks)
Hyperglycemia (require adjustment of diabetic medications)
- Mild to moderate hyperglycemia is common with thiazides and needs to be addressed.
- **Most thiazide diuretics are sulfa derivatives and can cause photosensitivity. Use sunscreen and wear protective clothing.**

Herbal Remedies

- ➔ Melatonin supplements
 - Help the body adjust quickly to new surroundings and time zones (jet lag) and fatigue.
 - Most practitioners agree that the lowest possible dose should be used and should be taken only for a short time.
 - High doses may cause side effects such as vivid dreams and nightmares.
 - Take melatonin once a person has reached the travel destination, as taking it before or during air travel can slow the recovery of jet lag, energy, and alertness.
- ➔ Evening primrose
 - May be used for eczema or skin irritations
- ➔ Ginseng

- Used to promote mental alertness and enhance the immune system.

➔ St. John's Wort
-used for treating depression.

DKA

- Life threatening emergency caused by a relative or absolute insulin deficiency.
- Hyperglycemia, ketosis, metabolic acidosis, and dehydration.
- Contributing factors: stress associated with illness/infection (elevated temp); and inadequate insulin dosage and self-management.
- Deficient fluid volume due to osmotic diuresis is priority (dehydration electrolyte imbalance, possible hypovolemic shock and renal failure).

Enoxaparin (levenoX)

- Low molecular weight heparin (LMWH) may be prescribed for 10-14 days following hip/knee surgery to prevent DVTs.
- Pinch skin upwards, insert needle 90 degrees into fold of skin.
- Mild pain, bruising, and irritation or redness of the skin at the injection site is common. Do NOT rub the site with the hand. Using an ice cube on the site can provide relief.
- Avoid taking aspirin or NSAIDS (ibuprofen) and herbal supplements (ginkgo, biloba, vitamin E), without HCP approval as it can increase the risk of bleeding.
- Vit K rich foods do not need to be eliminated from the diet during this drug therapy. PT and INR are not affected
- Monitor complete blood count (CBC) to assess for thrombocytopenia
- Routine coagulation studies (PT, INR, PTT, do not need to be monitored with enoxaparin)

Acetaminophen can damage the liver if taken too much.

Ibuprofen can cause GI/stomach upset and internal bleeding

PTT checks Heparin.

PT checks warfarin. INR standardizes the PT test.

Splenectomy

- Part of the immune system and functions as a filter to purify the blood and remove specific microorganisms that cause infections (pneumonia, meningococcal meningitis)

- Bacterial infection and rapid-onset sepsis are major lifelong complications in a client without a functioning spleen. A minor infection can become life threatening (fever, chills, headache are s/s). They need immediate intervention (cultures, imaging, a/b therapy).

MRI

- Contraindications: aneurysm clips, metallic implants such as ICDs, pacemakers, electronic devices, hearing aids, and shrapnel.
- Non-invasive test that does not require anesthesia or the use of iodine.
- **Gadolinium contrast is used instead**
- Loud noise and claustrophobia

Disorders of the aortic valve are best auscultated over the aortic area

- Right sternal border, second intercostals space

Steps to assess the abdomen

- Stand at client's RIGHT side; inspect; auscultate; Percuss; palpate

Fluoxetine (Prozac)

- Antidepressant

Lithium

- Mood stabilizer to treat bipolar disorder.
- Drowsiness, weight gain, dry mouth, and GI upset are expected, mild s/e.
- Toxicity: due to dehydration, low sodium diet, decreased renal function, and drug-drug interactions (NSAIDS, thiazide diuretics).
- Excessive urination and Polydipsia indicate nephrogenic diabetes insipidus from lithium toxicity.
- Dehydration and sodium loss from diarrhea and vomiting can lead to toxic lithium levels.

Incidence reports

- Adverse events are injuries caused by medical management rather than a client's underlying condition.
- Types of errors: diagnostic, preventive, treatment, and failure of communication,

equipment, or other systems.

Digoxin

- Cardiac glycoside that increases cardiac contractility but slows the HR and conduction.
- It is used in HF (to increase cardiac output) and atrial fib (to reduce the HR).
- The drug is excreted almost exclusively by the kidney. BUN and Cr levels are measurements of kidney function, and early signs of digoxin toxicity are N and V. Late signs are arrhythmias, including heart blocks. Monitor frequently.
- Take apical HR for 1 min prior to administration. It's safe to give if it's ≥ 60 /min.
- Hypokalemia can increase the risk of toxicity.

Incidence reports

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Potentially harmful drugs to avoid or administer with caution in the elderly d/t high incidence of drug toxicity, cognitive dysfunction and falls:

- Antipsychotics
- Anticholinergics
- Antihistamines
- Antihypertensives
- Benzodiazepines
- Diuretics
- Opioids
- Sliding insulin scales
- Amitriptyline is an antidepressant used to treat depression and neuropathic pain. It's anticholinergic properties can cause dry mouth, constipation, blurred vision, and dysrhythmias
- Chlorpheniramine is a sedating histamine H1 antagonist used to treat allergy symptoms. It can increase CNS effects (drowsiness, dizziness)
- Lorazepam (Ativan) is a benzodiazepine (tranquillizer, hypnotic, muscle relaxant) with a long half life (10-17hours) s/e: drowsiness, dizziness, ataxia, and confusion.
- Benzos: antipsychotics (valium, clonazepam)

Infants and hypoglycemia

- Infants of diabetic mothers are at risk of hypoglycemia and hypocalcemia.
- Hypoglycemia is considered a blood glucose level <40 mg/dL.
- Symptoms: jitteriness, irritability, apnea, lethargy, and temperature instability.

Infants

- Normal head circumference is between 32-37 cm
- Jaundice in the first 24 hours is not normal and it should be investigated.
- Should void and pass meconium in the first 24 hours after delivery
- Nasal flaring, chest wall retractions, and grunting with respirations are signs of distress

Anticholinergics

- Blocks the neurotransmitter acetylcholine in the CNS and PNS. Therapeutically, these drugs are used to relax muscles and dry secretions.
- Common indications:
 - Dicyclomine (bentyl) for symptomatic relief of GI hypermotility or IBS
 - Oxybutynin (Ditropan) for overactive bladder
 - Atropine as a mydriatic eye drop to cause dilation for refraction testing, and to dry respiratory secretions in end of life care.
- Contraindicated in glaucoma where it would affect intraocular pressure, bladder retention (BPH) and bowel obstruction/ileus
-

Codeine

- An opioid drug to treat mild to moderate pain, and also an antitussive to suppress the cough reflex
- Common adverse effects: constipation, N, V, orthostatic hypotension, dizziness
- Decreases GI motility, resulting in constipation. Need to increase fluids and fiber in diet and take laxatives to prevent constipation.
- Taking this with food helps prevent GI irritation (N, V)
- Photosensitivity, insomnia, palpitations, and anxiety are not adverse effects associated with codeine.

Types of shock

- Anaphylactic shock
- Hypovolemic shock
- Cardiogenic shock
- Septic shock
- Normal PAWP (pulmonary artery wedge pressure) is 6-12 mm Hg.

Kidney transplant patients

- Prescribed immunosuppressants (cyclosporine, azathioprine, prednisone) to help prevent

organ rejection.

- This can increase risk for developing infection. nurse should notify HCP of any signs of infection and abnormal urinalysis.

Vital signs

- Liver is a highly vascular organ and bleeding is a major complication. If tachycardia is present it can be an early sign of internal hemorrhage. This client should be assessed.

Diltiazam

- CCB, to get the heart rate <100, used for atrial fib and other cardiac conditions, HTN, raynaud's disease.

Methadone

- Potent narcotic and is unique in its long half life (up to 50+ hours) due to its lipophilic properties. There is a risk of overdose as the analgesic effect only lasts 6-8 hours and so a client can inadvertently take too many tablets for additional pain relief.
- Early signs of toxicity: N, V, and lethargy. Falling asleep with stimulation is classified as obtunded and requires additional observation/monitoring. Sedation precedes respiratory depression, which is a life threatening complication of severe toxicity.
- Normal healthy adult should have a pulse oximetry reading of 95-100%. Anything lower indicates an inadequate depth or rate of respiration.
- Itching sensation with narcotics is expected. It can be managed with an antihistamine.
- Occasional premature ventricular contractions are common in most adults. With methadone use, there is a risk of prolongation of the QT interval, so the client should have cardiac monitoring. However, in the absence of a prolongation, it is safe to discharge client.
- Torsades de pointes can also occur.

RN

- Cannot delegate assessment, teaching, evaluation, and clinical judgment to the LPN.

Retinal detachment

Can be caused by aging or head trauma (MVAs, etc)

- Flashes of light, floaters or black spots across the visual field, the sense of a curtain being drawn over the eye and loss of a portion of the visual field.
- This is an emergency and will lead to blindness if not treated.
- Primary intervention is to cover both eyes with patches to prevent further detachment.
- Place client on bed rest, notify HCP or ophthalmology specialist, perform visual acuity exam, and make client NPO for possible emergency surgery.

SPRAIN

- Stretch and/or tear of a ligament.
- Treatment for a sprained ankle:

REST, ICE, ELEVATE

Rest- activity stopped and movement limited for 24-48 hours to promote healing. No weight bearing on the joint for 48 hours, and crutches may be required.

Ice- cold, cryotherapy. Cold and ice therapy should be applied for 10-15 mins every hour for the first 24-48 hours. Vasoconstriction helps reduce pain, inflammation, and swelling.

Ice should not be applied directly to the skin. After the first 24-48 hours, moist heat can be applied for 20-30 mins at a time to reduce swelling with cooldowns in between.

Compression (ACE, wrap, splint)- pressure/compression can help prevent edema and promote fluid return.

Elevation- elevate the extremity above the heart on pillows for 24-48 hours to help reduce swelling by promoting fluid return.

Analgesia- mild analgesia with NSAIDs (ibuprofen) can be taken Q6H as needed to relieve pain and reduce swelling.

Exercise rehab program- this should be initiated as soon as possible after the injury (when pain subsides) to restore range of motion, flexibility and strength and prevent re-injury.

WILMS TUMOR

- Kidney tumor that usually occurs in children age <5.
- Often diagnosed after caregiver observe an unusually contour in the child's abdomen.
- Once diagnosis is suspected and confirmed, the abdomen should not be palpated, as this can disrupt the tumor. Post a sign that says "NO NOT PALPATE ABDOMEN" at bedside. Handle child carefully during bathing.

DROPLET PRECAUTIONS

N meningitides

H influenza type B

Diphtheria

Mumps

Rubella

Pertussis

Streptococcus group A (strep throat)

Viral influenza

PPE:

Surgical mask, private room, and as needed for procedures with risk of splash or body fluid contact: nitrile gloves, disposable gown, and goggles/shield.

Modafinil

- Stimulant and wakefulness promoting drug that is used for daytime sleepiness, narcolepsy, etc

Third Spacing

- Can occur following extensive abdominal surgery and can lead to hypovolemia, decreased CO, hypotension and tachycardia, and decreased urine output.
- Can occur 24-72 as a result of increased capillary permeability due to tissue trauma. It occurs when too much fluid moves from the intravascular into the interstitial or third space, a place between cells where fluid does not normally collect (injured site, peritoneal cavity).
- Priority is to assess vitals, weight gain, decreased urine output, hypovolemia, tachycardia, and hypotension.
- If not recognized earlier, it can lead to decreased renal perfusion, prerenal failure, and hypovolemic shock.

Peripheral IV catheter sites

- Should be changed every 72 hours or according to hospital policy, or if signs of complications develop, or if the patient requests it.
- Assess frequently for signs of infiltration, phlebitis, or infection (erythema, edema, warmth, pain, coolness to touch, palpable venous cord, edema).

Tumor Lysis syndrome

- A complication of chemotherapy is acute tumor lysis syndrome (TLS).
- It's a rapid release of intracellular components into the bloodstream. Massive cell lysis releases intracellular ions (K, Phos) and nucleic acids into the bloodstream.
- Catabolism of the nucleic acids produce uric acid, resulting in severe Hyperuricemia.
- Released phosphorus binds calcium, producing calcium phosphate mixture but lowering serum calcium levels. Both calcium phos. And uric acid are deposited into the kidneys, causing renal injury.
- Allopurinol (zyloprim) blocks the nucleic acid catabolism and prevents Hyperuricemia but would not affect K, phos, and calcium levels.
- Cancer chemo can cause cell lysis, due to massive release of nucleic acid and its metabolic product, uric acid. Uric acid deposition leads to acute kidney injury. Allopurinol or rasburicase, and aggressive IV hydration are used to prevent this complication.
- Normal blood uric acid levels for adults: 4.0-8.5 md/dL

Rheumatic fever

- Noncontagious acute fever marked with inflammation and pain in joints d/t strep. Infection. it occurs in young people.

Rheumatic heart disease

- Permanenet heart damage following rheumatic fever, can lead to HF.

Multiple Sclerosis

- Disease of the SNS, brain and spinal cord, where the insulating covers of the nerve cells are damaged.

Cystic Fibrosis

- Pulmonary complications due to thick mucus that trap bacteria. respiratory infections are lifelong issues.

Cyanocobalamin

- Used for B12 deficiency, and usually administered Q4weeks

Apical HR

- Mitral/apex area on the 5th ICS on the MCL

Filgrastim (neupogen) and Pegafilgrastim (Neulasta)

- Used to treat Neutropenia and stimulate the bone marrow to increase its production.
 - Used during chemotherapy
 - Chemotherapy can cause suppression of rapidly reproducing cells, including bone marrow suppression. This leads to decreased RBCs, WBCs, and platelets.
 - Leukopenia is decreased WBCs (<4000) and Neutropenia is decreased neutrophils (<1500)
-

Malignant hyperthermia

- Life-threatening inherited muscle abnormality that is triggered by certain drugs used to induce general anesthesia.
- Leads to excessive release of calcium from the muscles, leading to sustained muscle contraction and rigidity.
- It can occur in the OU or in the PACU.
- s/s of MH: hypercapnia (earliest sign), general muscle rigidity (jaw, trunk, extremities) and hyperthermia (late sign).
- Monitor temp., it can rise 1 c every 5 mins and can exceed 40.6C
- Notify HCP, give dantolene (muscle relaxant), cooling blanket, fluid resuscitation.

Hypertensive crisis

- Life threatening emergency issue due to possible organ damage.
- Can lead to intracranial hemorrhage, HF, MI, renal failure, aortic dissection, or retinopathy

- Treatment: IV vasodilators (nitroprusside sodium). Lower BP slowly, too rapid a drop can cause decreased brain, heart, kidney perfusion. Can lead to stroke, etc.
- Decrease MAP by no more than 25% or maintain MAP at 110-115 mm Hg
- $MAP = (2 \times DBP + SBP) / 3$

Radiation

- Damages the DNA, causes cell destruction.
- Radiation and chemo affects tissues with rapidly proliferating cells (oral, mucosal, GI, bone marrow) first followed by tissues with slowly proliferating cells (cartilage, bone, kidney).
- Early damage s/s: oral mucosal ulcerations, V, D, low blood cell counts.

Neurogenic shock

- Hypotension and Bradycardia
- Causes disruption of the function of the SNS not the PNS

NORMAL

- CVP: 2-8 mm Hg
- MAP: 70-105 mm Hg
- SVR: 800-1200 dynes/sec/cm-5

Synchronizing Cardioversion

- Delivers shock to the R wave of the QRS complex.
- Rhythms ideal for the synchronized Cardioversion: supraventricular tachycardia, ventricular tachycardia with a pulse, and atrial fib with rapid ventricular response

Neurogenic shock with spinal injury

- Vascular dilation with decreased venous return to the heart, due to loss of innervations to the spine.
- Hypotension, Bradycardia, pink and dry skin d/t vasodilation.
- Neurogenic shock usually occurs in cervical or high thoracic injuries (T6 or higher)
- Systolic BP should remain at 80 mm Hg or above to adequately perfuse the kidneys.
- Administering fluids is top priority to ensure adequate kidney and other organ perfusion.

Pulmonary artery catheter

- Measures the right ventricular preload and reflects the fluid vol. Status. It is used to measure the CVP.
- Use the proximal port/lumen to measure this, as its lumen is located in the right atrium.
- Normal CVP: 2-8 mm Hg
- Low CVP= hypovolemia
- High CVP= right vent. Failure or fluid volume excess.
- CVP is measured as a mean pressure and should be recorded at the end of expiration.

Basic Life Support after assessing responsiveness (unconscious and unresponsive client):

- Verify loss of consciousness with a sternal rub; check breathing for 5-10 seconds
- Activate the Emergency response system by calling 911 or code, get AED
- Check carotid pulse 5-10 seconds to verify cardiac arrest
- Attempt CPR if no pulse
- Notify HCP

VACCINES

- Tdap given to pregnant women ideally between 27th and the end of the 36th week of gestation.
- Influenza inactivated IM can be given during pregnancy from Oct-March
- Influenza nasal spray is a live vaccine, given to those between 2-49 y/o. These are not safe for pregnant women as it is harmful to the development of the fetus and embryo.
- MMR is live and is not safe for pregnant women d/t risk of birth defects and miscarriage from rubella component.
- Varicella (chickenpox) is a category X, live, attenuated vaccine that is contraindicated for pregnancy or a planned pregnancy within 4 weeks of administration. Can be given after birth to a woman who is breastfeeding her infant.

Gestation

- NSAIDS (ibuprofen, indomethacin or naproxen) inhibit uterine contractions that occur d/t prostaglandin synthesis, and can be taken to decrease pain and inflammation or reduce fever.
- Ibuprofen- Pregnancy category C through 29 weeks gestation and pregnancy category D starting at 30 weeks gestation. Must be avoided after 30, d/t risk of premature closure of the ductus arteriosus in fetus and prolonged labor.

Interventions for back/sciatic pain during pregnancy:

- Acetaminophen.
- Heat: warm bath or warm compress to relieve discomfort
- Reposition: lying on the unaffected side or changing positions may help relieve pain.

Metronidazole (Flagyl)

- An anti-infective
- Might cause urine to turn dark

DEMENTIA

Drugs:

→ Aripiprazole (abilify)

- Atypical antipsychotic that acts as a dopamine system stabilizer.
- Stabilizes mood and control symptoms such as agitation and hallucinations in clients with dementia.

→ Donepezil (Aricept)

- Used to prevent worsening symptoms in Alzheimer Dementia, and it does not decrease agitation. It's a preventive med and would be used as second priority.

Lasix and ACE inhibitors are usually taken together to control HF symptoms, esp with those with fluid overload.

Calcium carbonate

- Used for osteoporosis , it's a calcium supplement.

Antiemetics

- Ondansetron (zofran)
- It helps prevent straining that can cause wound dehiscence.

AFRICAN AMERICANS

→Cervical Cancer

- Incidence highest among Hispanics, American Indians, and African Americans.
- Mortality rate among African American women for cervical cancer is 2X higher than that for white women

→Hypertension

- African Americans have the highest incidence in the world, and more common in women than men.
- The mortality rate for women in this group is higher than the white.

→Ischemic stroke

- Risk factors are related to increased HTN, DM, and sickle cell anemia.

OSTEO

- Affects white and Asian women more, but affects all ethnic groups

Melanoma (skin cancer)

- More common in white, light-skinned, and over the age of 60 with frequent sun exposure.
- 10X higher incidence rate than African Americans

ACE Inhibitors

- Cause angioedema (rapid swelling of the lips, tongue, throat, face, and larynx).
- It can lead to airway obstruction and possible death if not treated immediately.
- Risk is 5X greater in African Americans than in Caucasians.
- d/c drug and notify immediately
- Common adverse effects: persistent dry cough d/t the buildup of bradykinin in the lung; orthostatic hypotension (dizzy) and hyperkalemia).

ADENOSINE

- Antiarrhythmic and anucleoside. Slows the electric conduction of the heart, slows HR., or normalizes heart rhythm.
- Used to treat irregular heart beats.
- Used during stress tests by improving the blood flow to the heart.
- Do not give:

During 2nd or 3rd degree heart blocks if no pacemaker

During breathing problems (asthma)

Allergic to any adenosine ingredients

- Treats rapid tachycardic rhythms such as PSVT

Buerger's disease (thromboangiitis obliterans)

- Nonatherosclerotic vasculitis involving the arteries and veins of the lower and upper extremities.
- Occurs most often in young men (<45) with history of tobacco or marijuana use and chronic periodontal infection, but no other CV risk factor.
- Thrombus formation, distal extremity ischemia, ischemic digit ulcers, or digit gangrene.
- Intermittent claudication of feet and hands.
- Over time, rest pain and ischemic ulcerations may occur.
- Some develop secondary Raynaud phenomenon (cold sensitivity)
- Stop all tobacco and marijuana use. Do NOT use nicotine replacement products.
- **Bupropion and varenicline can be used for smoking cessation.**
- Avoid cold exposure to limbs to prevent vasoconstriction and worsening of symptoms,, start a walking program, a/b for infected ulcers, analgesics for ischemic pain, and avoidance of trauma to the extremities.

Infective Endocarditis

- Vegetations over the valves can break off and embolize to various organs, resulting in life threatening complications:
 - stroke: paralysis on one side
 - Spinal cord ischemia: paralysis of both legs
 - ischemia to the extremities: pain, pallor, and cold foot or arm
 - intestinal infarction: abdo pain
 - splenic infarction: left upper quadrant pain
- If any of these symptoms occur, report to the HCP stat.
- Fever, arthralgia (multiple joint pains), weakness, and fatigue are expected and don't need to be reported.
- A/bs needed for 4-6 weeks. Fever may persist for days after treatment, but if the client is persistently febrile after 1-2 weeks of a/bs, report it (ineffective a/b therapy).
- Splinter hemorrhages are common with IE, d/t vessel damage from swelling of the blood vessels (vasculitis) or tiny clots that damage the small capillaries (microemboli).

HEART RHYTHMS

→Supraventricular tachycardia

- It's a **regular narrow-complex QRS tachycardia**
- Dysrhythmias originates from ectopic focus above the bifurcation of the bundle of His.
- **Heart rate can be 150-220/min.**
- Regular rhythm.
- **P waves hidden often**, if visible, it has an abnormal shape and the
- **PR interval may be shortened.**
- **The QRS complex is usually narrow (<0.12 second).**
- **Retrograde P waves (pseudo S waves) following QRS**
- Stimulants (nicotine, caffeine, cocaine) and organic heart disease can cause SVT.
- A prolonged episode of SVT with a HR >180 will cause decreased CO and hypotension, palpitations, dyspnea, and angina.
- Treatment: vagal stimulation (Valsalva, coughing, and carotid massage);
- **drug therapy (IV adenosine is the DOC)**. If this doesn't work and the client is unstable, synchronized Cardioversion is used. Recurrent SVT may need radiofrequency catheter ablation.

→ Atrial Fib

- **P waves not present, fibrillatory waves present.**
- **Rhythm is usually irregularly irregular** (the R waves).

→ Sinus Tachy

- **HR: 101-200/min**
- **Normal P wave preceding each QRS interval**, normal shape and duration.
- **The PR interval is normal** (0.10-0.20 seconds which is <5 small squares)
- **QRS is <0.12 seconds** (<3 small squares).

→ Ventricular tach

- Vent rate: **150-250/min** and originates from foci firing in the ventricle repeatedly.
- **P waves are usually not visible** and are buried in the QRS
- **PR interval is not measurable.**
- **The QRS complex is typically wide** (>0.12 seconds)

ARTHRITIS

→ rheumatoid arthritis

- Chronic, symmetrical, bilateral, autoimmune, inflammatory pain and swelling in the small joints of the hands.
- Unknown origin, periods of exacerbations and remission.

- No cure
- Morning joint stiffness that lasts 60 mins-several hours
- Elevated ESR and rheumatoid factor levels
- Can involve the cervical spine

→ osteoarthritis

- Degenerative joint disease of the synovial joints (knee, hip, fingers).
- Progressive erosion, asymmetrical.
- Most common form of arthritis in >55 years
- Aching pain, exacerbated by muscle bearing activities, causing synovial inflammation, muscle spasm, and nerve irritation.
- Crepitus, a grating noise that is heard or palpated d/t bone or cartilage fragments floating.
- Morning stiffness that subsides within 30 mins of arising.
- Decreased joint mobility and ROM
- Atrophy of the muscles that support the joint (quads, hamstring) due to disuse.
- Typically noninflammatory, nonsystemic disorder. Therefore low grade temp is not an s/s.

lactose intolerant

→ Calcium sources

- Some fish (sardines, salmon, trout)
- Tofu
- Some greens (spinach, kale, broccoli)
- Almonds

→ Vit D sources

- Egg yolk and oily fish (salmon, sardines, tuna)

LATEX ALLERGY

- Cross allergy foods can cause latex food syndrome: bananas, kiwi, avocados, tomatoes, peaches, grapes, as some proteins in rubber are similar to food protein.
- Allergic contact dermatitis (rash, itching, vesicles)- develop 3-4 days after exposure, a type IV hypersensitivity reaction (delayed onset)
- Anaphylaxis is a type 1 hypersensitivity reaction and should be treated with IM epi injections. (gloves, balloon tip catheters, blowing up toy balloons, use of bottle nipples, pacifiers, condoms or diaphragms during sex)
- Carry epi pen and wear a medic bracelet

- 73% with spina bifida have a sensitivity to latex.
- Screening questions: do lips swell after blowing up balloons (which have latex)? Do your hands itch and/or burn after wearing rubber gloves?

Sjogren's syndrome

- A chronic autoimmune disorder in which moisture-producing exocrine glands of the body are attacked by WBCs.
- Salivary, lacrimal glands are most affected, leading to dry eyes, dry mouth. Dryness can lead to corneal ulcerations, dental caries and oral thrush.
- Skin (dry and rashes); throat and bronchi (chronic dry cough); vagina (dryness and painful intercourse) can also occur.
- Treatment: alleviate symptoms, there is no cure. OTC or prescribed droplets to relieve itching, burning, dryness, and gritty sensation in the eyes. Wearing goggles may offer further protection from drying caused by the wind. Dry mouth is treated with sugarless gum and candy or artificial saliva. Regular dental appointments to prevent caries are recommended. Lubricants help to ease vaginal dryness. Avoid low humidity environments (airplanes, central heated houses) and use humidifiers (mainly at night).
- Avoid decongestants as they can further dryness to the mouth and nasal mucosa.
- Avoid oral irritants (coffee, alcohol, nicotine)
- Avoid acid drinks (carbonated beverages, juices) and sip water instead

AIRBORN PRECAUTIONS

- Measles (rubeola), TB, varicella (chicken pox, herpes zoster), severe acute respiratory syndrome
- Are the most contagious of pathogens and they should be isolated first amongst other precautions.
- These travel through small air particles that circulate the air
- N95 particle respirator
- As needed during risk of splash or body fluid contact: Nitrile gloves, disposable gown, goggles/shield
- Negative air pressure room, high efficiency particulate air filters.
- MY COW HEZ TB

DROPLET PRECAUTIONS

- Pertussis infection
- Influenza
- PPE: surgical mask

CONTACT PRECAUTIONS

- MRSA, C. Diff, VRE, scabies, draining abscess, head lice, pink eye, diarrhea in young child, herpes simplex (open lesions), impetigo
- PPE: disposable gown, nitrile (regular, clean) gloves
- Hand hygiene: with soap and water; alcohol based sanitizers don't kill C diff spores.

REMOVING PPE

Gloves, face shield/goggles, gown, mask/respirator

Medic alert bracelets

- Generally worn by the following:
Epilepsy, anaphylactic allergies, diabetes, dementia, or hemophilia

NORMAL INR

- 0.75-1.25

Cardiac Catheterization Complications:

- Contrast dye is used. Check for allergies to shellfish, iodine, etc.
 - Pre medicate client with corticosteroids and H1 antihistamines
 - Contrast nephropathy: contrast can cause kidney injury, which can be prevented with adequate hydration. Contrast use should be restricted in clients with renal disease unless needed for a life-saving procedure.
 - Metformin (glucophage) given with large-dose IV iodine contrast can increase the risk for lactic acidosis. As a result, most clinicians discontinue metformin on the day of IV contrast exposure (regardless of baseline creatinine) and restart the drug at least 48 hours later after documenting stable renal function.
 - Glucose metabolized, leads to lactate formation
- ➔ Cardiac catheterization involves the use of iodinated contrast. Contrast can induce allergic reactions, lactic acidosis, and kidney injury. Clients cannot have the test if they have taken metformin that day, have allergies to the dye (shellfish) or have kidney disease.

SEROTONIN

- There must be a minimum of 14 days between the administration of MAOIs and SSRIs to avoid serotonin syndrome, and these meds cannot be administered together.
- **SSRIs:** used as antidepressants in the treatment of major depressive disorder and anxiety disorders. (**escitalopram**)
- **MAOIs (phenelzine):** treat atypical depression

Shingles

Herpes zoster; a reactivation of the varicella zoster (chicken pox) virus.

Most likely to occur in immunocompromised clients or treatments

Lesions that are open, may transmit both by air and contact.

DISSEMINATED (spread) shingles that are not crusted over will require contact, airborne, and negative airflow room to prevent transmission.

Localized shingles require only standard precautions for clients with intact immune systems and contained/covered lesions.

ALLERGIES

- Iodine/shellfish: during CT scans and cardiac catheterization procedures
- Gadolinium (non-iodine contrast material): during MRI, MRCP

URINE SPECIMENS

- Are collected aseptically from the port located on the catheter tubing on an indwelling urinary catheter. The collected urine should be measured and discarded as collected urine in the bag may lead to incorrect results.
- Clean the collection port with alcohol swab, aspirate urine with a sterile syringe, use aseptic technique to transfer the specimen to a sterile specimen cup.
- Midstream and clean catch methods are not necessary if a catheter is already in place.
- Specimens should be kept cool until transported to the lab.
- Never take a urine specimen from the collection bag.

IV GAUGE CATHETERS

- A lower IV catheter gauge number corresponds to a larger bore IV catheter.

- 14 gauge (large bore) catheter may be used for administering fluids and drugs in an emergency or prehospital setting, or for hypovolemic shock.
- 18 gauge: in somewhat stable adult clients who require large amounts of fluids or blood, this size is recommended.
- 20-22: sufficient for administering general IV fluids and meds to adult clients. **This is not preferred for administering blood.**
- 24: recommended for children and some older adults with small, fragile veins.

Written consent

- Required for invasive procedures and surgery.
- Nurses role: to WITNESS that the client signed the consent voluntarily and was COMPETENT at the time of signing. Ensure that the client received necessary info and does not have remaining questions about the procedure. Document in the client's records that the informed consent was given and the date/time of the signature.
- HCPs role: explaining all aspects of the procedure, ensuring that the client has a correct understanding of the procedure/risks, providing names/qualifications of those who will be involved, describing available alternate treatments, and reinforcing that the client has the right to refuse the procedure.